UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG

SCHOOL OF MECHANICAL, INDUSTRIAL & AERONAUTICAL ENGINEERING

Aggressive Manoeuvre Stabilisation of Agricultural Drone

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EXECUTIVE SUMMARY

The Fourth Industrial Revolution has made the implementation of drones in the agriculture sector more common, particularly in precision farming. This type of farming requires the drone to not only perform basic operations such as hovering but necessitates an ability to stabilise after free throws and free falls as well as external interferences such as strong wind forces and collisions with crops and birds. Preliminary analysis by means of analytical and numerical Bode and Nyquist plots proved the uncontrolled model to be unstable. As such, a control model was implemented to achieve drone stability. This was achieved using analytical calculations based on the linearised model as well as MATLAB and Simulink analysis on both linear and nonlinear models. Initially, the control considered extended to proportional control, integral control and PI control, however, these failed to effectively stabilise the drone. Proportional control provided marginal stability and Bounded Input Bounded Output (BIBO) instability and hence could not be implemented in isolation. Additionally, the lack of damping evident when using the integral control and PI control meant that stability could not be achieved according to the Routh Hurwitz criterion. PD and PID control both provided drone stability as both methods had a damping term. However, the integral control evident in PID created a third order transfer function which could not effectively be solved using the second order solving techniques available. Thus, the PD controller was selected for the robustness analysis performed on the drone. Lastly, it was determined that the drone must be fitted with a microcontroller, altimeter, IMU and GPS model such that the control model could operate.

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1 INTRODUCTION

The global agricultural industry continuously encounters increasing pressures due to rapid population growth, climate change and limited resources that directly influence the urgent need for sustainable food production [1]. These pressures have urged a technological shift within the Fourth Industrial Revolution toward the integration of data-driven automated systems – specifically, precision agriculture – over traditional methods to meet the rising demand for food security. Precision agriculture leverages advanced technologies that incorporate information systems to develop crop production optimization techniques that both reduce resource consumption and reinforce environmental sustainability [2].

At the forefront of these techniques, unmanned aerial vehicles (UAVs), commonly known as drones, have been rendered crucial to driving this shift. Originally developed for military observation, UAVs have evolved to versatile tools used in everyday life for recreation, surveying, logistics and emergency responses [3]. Within precision agriculture, drones are deployed for tasks such as crop spraying, soil analysis and aerial surveillance [4]. The drones are equipped with sophisticated sensors such as multispectral, Light Detection and Ranging (LiDAR), thermal and RGB cameras to obtain detailed evaluations of field mapping, crop health, soil conditions, water distribution and pest infestations with precision [5]. Furthermore, their ability to capture high-resolution images at low altitudes and reach challenging areas cause them to be significantly useful in managing large fields and complex terrains.

Drones allow farmers to make real-time decisions using data from advanced sensors, a process that was hindered by traditional monitoring methods. Traditional monitoring includes field scouting and satellite imagery that often suffer from delays in data acquisition. This is due to inadequate spatial monitoring either through limited manpower for effective ground coverage or low-resolution satellite images that is often blurred by cloud coverage [6]. These constraints induce high labour and operational costs for farmers. Consequently, drones have gained popularity as an economically viable solution for optimizing timely crop management.

However, the advantages of UAVs within precision agriculture come with significant challenges during operation. The drone experiences unpredictable disturbances within its environment that complicate its flight trajectory and data acquisition. This may be due to wind gusts, drone handling or sudden impacts. Moreover, the sensors contribute to the drone's centre of gravity and moment of inertia, which must be accounted for within the system model as it may disturb the system's robustness. These considerations highlight the necessity of a control system to maintain stability and trajectory accuracy under real-world conditions.

1.1 Scenario Description

Quadrotor drones are widely used for agricultural surveys due to their ability to develop real-time maps to monitor and assess crop conditions across fields. The following scenario is developed to analyse the control of an agricultural surveying quadrotor:

Sugar cane is a vital plant within the agricultural economy of South Africa. A smallholder farmer in Kwa-Zulu Natal, oversees a 2-hectare sugar cane plantation. The farm maintenance demands weekly scouting to detect

infections, infestations, soil deficiencies and water clogging [7]. However, traditional crop scouting relies on ground-based observations. This poses challenges as the sugar cane grows up to 3 metres tall and has minimal gaps between plants that obscure the scout's visibility and access to view stem thinning or disease areas on crops [8, 9]. Figure 1 below showcases a visual representation of the conditions that scouts are subjected to. These factors, combined with the high labour costs to perform scouting, often lead to delayed disease detection that result in reduced yields [10].



Figure 1: Typical Sugar Cane Farm View in KwaZulu Natal, South Africa *[11]*

To overcome these limitations, the farmer utilizes a quadrotor drone to perform crop inspections. Their simple structure, compact design, high manoeuvrability and vertical take-off and landing capabilities enable agile flight in versatile environments [12]. Flying above the crops provides a bird's-eye view that ground scouting cannot, this enables faster crop evaluation for the farmer to implement immediate preventative measures for detected diseased plants.

The drone utilized in this scenario is the DJI Mavic 3M, as seen in Figure 2, designed for agricultural aerial surveying. Its combined multispectral and RGB camera allows high resolution scanning of crops to analyse crop growth and identify any concerning plant or soil features [13].



Figure 2: DJI Mavic 3M *[14]*

A system model must be determined to mimic the drone behaviour to be analysed. The mass and polar moment of inertia for the drone is calculated by obtaining the mass of the drone and making the following simplifying assumptions:

- Mass is uniformly distributed across the drone

- The drone is simplified to two simple beams placed perpendicularly; thus, symmetry is induced and therefore the moments of inertia about the x- and y- axes are equal
- The yaw, roll and pitch moments of inertia is calculated as follows:

$$I_{zz} = \frac{1}{3} \times m \times \frac{L^2}{2} \tag{1.1}$$

where L is the diagonal length of the drone.

$$I_{xx,yy} = \frac{1}{3} \times m \times \left(\frac{Lcos(45)}{2}\right)^2 \tag{1.2}$$

where L is the distance from the centre of mass of the drone.

Table 1: DJI Mavic 3M Specifications

Drone Parameter	Value
Mass [15]	1.05 kg
Assumed Length of Drone	238 mm
$I_{xx,yy}$	0.002494 kg·m²
I_{zz}	0.004988 kg·m²

A key feature the DJI Mavic 3M must achieve is stable hovering, achieved when the rotors' thrust balance the gravitational forces [12]. Hovering is essential for close range crop inspections to ensure high resolution data is captured with accuracy, even in the presence of external disturbances. To navigate the tightly spaced, towering crops and gather comprehensive information, the drone must retain its hovering position upon experiencing aggressive manoeuvres. Effective control systems are therefore critical to maintain the quadrotor's stability in dynamic conditions.

1.2 Aggressive Manoeuvre Inputs

To conduct effective agricultural surveys, the drone must execute aggressive manoeuvres and withstand external disturbances. In the fast-paced environment of a smallholder sugar cane farm where time and resources are limited, farmers may launch drones via a freehand throw to quickly deploy the drone over the field. The drone must rapidly stabilize its position from the unpredictable velocities and orientations to shift from chaotic motion to controlled flight. Without control, the drone will fail to hover and risks drifting into crops, damaging both the quadrotor and sugar cane.

During an aerial survey, the drone is also subject to operator error whereby the input command triggers an uncontrollable free fall, once again risking damage to the equipment and crops. Thus, the control system must rapidly detect the fall and regain the balance between thrust and the system's weight before impact.

Within the environment of a sugar cane farm, the drone may experience collisions with cane stalks and leaves during low altitude flight for detailed plant inspections. These collisions may snag the drone propellers causing

the drone to flip upside down. The farmer attempting sharp turns to avoid irrigation lines can also be considered to be an aggressive manoeuvre that causes the drone's reorientation. The control system must resettle the drone to its upright position to avoid further collision with crops. The drone's uncontrolled attitude must be restored to stable hovering.

KwaZulu Natal's coastal climate has unpredictable wind gusts that affect farm operations. The 10% windiest areas of the province experience annual average wind speed of 4.91 m/s (17.6 km/h) at a height of 10 m [16]. The gusts can disrupt the flight trajectory or tilt the drone while it hovers during crop assessments. These challenges require the controller to continuously adjust the thrust and orientation of the drone to counteract the wind forces experienced and remain hovering. Otherwise, the drone may drift and compromise the mapping accuracy during data collection.

Due to the sugar cane farm's rural placement, birds are attracted to the plantation due to the food sources that surround the fields [17]. As such, sudden collisions between birds and the drone should be compensated by the control system to quickly return to its stabilized hover state. These aggressive manoeuvres and external disturbances highlight the crucial requirement for drone to implement real-time feedback and adaptive control systems within its operation. Stable hovering not only mitigates the effects of these disturbances but is also necessary to produce high-quality data required from aerial surveying within precision agriculture.

2 DRONE SYSTEM MODELLING

A quadrotor operates with six degrees of freedom (DOF), controlled by four inputs that commands the rotor speeds, which generates the thrust and torques to manoeuvre the drone. The degrees of freedom consist of three translational movements (x, y, z positions) and three rotational movements (roll, pitch and yaw). This enables the drone to navigate and orient itself according to user-specified coordinates within the three-dimensional plane of the real world.

However, the system is considered underactuated as the number of control inputs is (four) is fewer than the DOFs (six). The nature of the underactuated system limits the drone's operation as the inputs cannot simultaneously control all six DOFs independently, this directly causes instability, necessitating the need for control [18].

To design a controller to mitigate the instability, especially during aggressive manoeuvres, a detailed mathematical model that represents the quadrotor's dynamics is required. The nonlinear model describes the drone's complete kinematic and dynamic behaviour. The linearized model approximates the nonlinear dynamics by simplifications in order to implement existing control theories to stabilize the drone [19].

The following assumptions are made to model the drone behaviour [20]:

- The quadrotor and its propellers do not deform during flight and are considered rigid
- The gyroscopic forces generated by the propellers and friction forces are neglected

2.1 Nonlinear Model

The nonlinear model is a true representation of the quadrotor's dynamics governed by Newton-Euler laws within two reference frames – Earth and body [20]. The Earth reference frame refers to the positioning of the drone in an inertial reference frame that is fixed to the ground. The body frame is the movement of the drone relative to a plane that is fixed to the drone's centre of mass.

2.1.1 Governing system equations

The governing system equation utilised in this model was obtained from *Different Linearisation Control Techniques for a Quadrotor System* by M. Belkheiri et al [21]. The matrix of the derivative of the state vectors in the mixed reference frame is provided below. In this equation, the positive y direction is taken as upwards.

$$\begin{bmatrix} \dot{\emptyset} \\ \dot{\theta} \\ \dot{\varphi} \\ \dot{z} \\ \dot{x} \\ \dot{y} \\ \ddot{\emptyset} \\ \ddot{\theta} \\ \ddot{\varphi} \\ \ddot{z} \\ \ddot{x} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} \dot{\emptyset} \\ \dot{\theta}\dot{\varphi}a_1 + \dot{\theta}a_2\Omega_r + b_1U_2 \\ \dot{\theta} \\ \dot{\emptyset}\dot{\varphi}a_3 + \dot{\emptyset}a_4\Omega_r + b_2U_3 \\ \dot{\varphi} \\ \dot{\theta}\dot{\emptyset}a_5 + b_3U_4 \\ \dot{z} \\ -g + (\cos\emptyset + \cos\theta)\frac{1}{m}U_1 \\ \dot{x} \\ U_x\frac{1}{m}U_1 \\ \dot{y} \\ U_y\frac{1}{m}U_1 \end{bmatrix} \tag{2.1}$$

Where $\emptyset, \theta$ and φ are roll, pitch and yaw respectively and x, y and z are translational positions. Additionally, the constants a and b, the rotor thrust and drag coefficients, are defined as:

$$a_1 = \frac{I_{yy} - I_{zz}}{I_{xx}} \tag{2.2}$$

$$a_2 = -\frac{J_r}{I_{xx}} \tag{2.3}$$

$$a_3 = \frac{I_{zz} - I_{xx}}{I_{yy}} \tag{2.4}$$

$$a_4 = \frac{J_r}{I_{yy}} \tag{2.5}$$

$$a_5 = \frac{I_{xx} - I_{yy}}{I_{zz}} \tag{2.6}$$

$$b_1 = \frac{1}{I_{xx}} \tag{2.7}$$

$$b_2 = \frac{1}{I_{yy}} \tag{2.8}$$

$$b_3 = \frac{1}{I_{zz}} \tag{2.9}$$

Lastly, U_1 to U_4 as well as U_x and U_y that describe the drone's thrust and moments are defined as:

$$U_1 = b\left(\Omega_1{}^2 + \Omega_2{}^2 + \Omega_3{}^2 + \Omega_4{}^2\right) \tag{2.10}$$

$$U_2 = lb\left(-\Omega_2{}^2 + \Omega_4{}^2\right) \tag{2.11}$$

$$U_3 = lb\left(\Omega_1{}^2 - \Omega_3{}^2\right) \tag{2.12}$$

$$U_4 = d\left(-\Omega_1{}^2 + \Omega_2{}^2 - \Omega_3{}^2 + \Omega_4{}^2\right) \tag{2.13}$$

$$U_x = \cos\emptyset \sin\theta \cos\varphi + \sin\emptyset \sin\varphi \tag{2.14}$$

$$U_y = \cos\emptyset \sin\theta \sin\varphi - \sin\emptyset \cos\varphi \tag{2.15}$$

where Ω represents the respective rotor speed.

2.2 Linearised Model

The nonlinear quadrotor model derived is essential for representing the general operating conditions. However, this model's nonlinearities and coupled terms do not allow the system to be analysed using well-established linear control techniques used to design control systems [22]. Thus, a linearised model of the system is necessary.

The linearisation of the system will utilize approximations of the nonlinear model around the steady hovering condition. Under this condition, disturbances and aggressive manoeuvres can be evaluated using simplified linear representation and the designed linear control system returns the drone to stable hovering.

2.2.1 Assumptions

The following assumptions were made to linearise the generalised equation described in Section 2.1.1:

- Generalised equation applied to the case of hovering only
- $\emptyset = \theta = \varphi = 0$
- $\dot{x} = \dot{y} = \dot{z} = 0$
- $\dot{\emptyset} = \dot{\theta} = \dot{\varphi} = 0$
- $U_2 = U_3 = U_4 = 0$
- $U_1 = mg$ (to maintain hovering height)
- Small angle approximations

2.2.2 Linearisation

Four states were considered when performing linearisation, namely $z, \emptyset, \theta, \varphi$. Four functions were created for the second derivatives for each of the states using the generalised equation in Section 2.1.1. These functions are defined as:

$$f_{10} + \ddot{z}_0 + g - \frac{U_{10}}{m} = 0 \tag{2.16}$$

$$f_{20} + \ddot{\emptyset}_0 - b_1 U_{20} = 0 \tag{2.17}$$

$$f_{30} + \ddot{\theta}_0 - b_2 U_{30} = 0 \tag{2.18}$$

$$f_{40} + \ddot{\varphi}_0 - b_3 U_{40} = 0 \tag{2.19}$$

To obtain the final linearised model, a Taylor series expansion was performed on each of the above equations at the equilibrium point. The terms of higher orders of magnitude were ignored due to their negligible effect on the final solution. Therefore, the linearised model obtained is shown in Equation 2.20 to 2.23 below.

$$-g + \Delta\ddot{z} - \frac{1}{m}\Delta U_1 = 0 \tag{2.20}$$

$$\Delta\ddot{\emptyset} - b_1\Delta U_2 = 0 \tag{2.21}$$

$$\Delta\ddot{\theta} - b_2\Delta U_3 = 0 \tag{2.22}$$

$$\Delta\ddot{\varphi} - b_3\Delta U_4 = 0 \tag{2.23}$$

2.3 Transfer Function

To implement the linear control techniques, the linearised model must be expressed as a transfer function. Transfer functions exhibit the input-output behaviour within the Laplace domain, assuming zero initial conditions for the Single-Input, Single-Output (SISO) [23]. The model input refers to the parameter that adjusts the drone thrust to control the drone's altitude and orientation.

Since four states were necessary to define the hovering of the drone, four transfer functions were obtained. This was done by setting all equilibrium conditions to zero, such that there are not delta terms. Subsequently, the Laplace Transforms of each equation was obtained and U_1 equalled *mg*.

$$\frac{Z(s)}{U_1(s)} = \frac{1}{ms^2} \tag{2.24}$$

$$\frac{\emptyset(s)}{U_2(s)} = \frac{1}{I_{xx}s^2} \tag{2.25}$$

$$\frac{\theta(s)}{U_3(s)} = \frac{1}{I_{yy}s^2} \tag{2.26}$$

$$\frac{\varphi(s)}{U_4(s)} = \frac{1}{I_{zz}s^2} \tag{2.27}$$

The transfer functions above depict the drone's dynamic response in the Laplace domain. This enables the application of well-versed control implementations as the double integrators provide a basis to conduct stability analyses. These analyses provide understanding of the linearised open loop system behaviour and its response and allow the control system to be designed using phase and gain margin insights to avoid loss of control as described by MATLAB's *Assessing Gain and Phase Margins* [24].

3 STABILITY ANALYSIS

It is essential to evaluate the system's stability behaviour before implementing a controller to understand the system responses to the drone inputs and disturbances. The stability analysis then becomes an assessment to determine whether the outputs are bounded, where the output remains in a finite range, or diverge over time [25].

The time- and frequency-domain responses are analysed to recognize instabilities within the system. Specifically, Bode and Nyquist plots are employed to evaluate the open loop linearised system. The Bode plots quantify the system's proximity to instability, thus determining safe operating ranges for the controller [25]. The Nyquist plot then provides a graphical representation of the system's frequency response within the complex plane. The Nyquist stability criterion is applied to evaluate whether a feedback loop upholds or collapses the system's stability [26].

3.1 Bode Plot

Bode plots represent the phase and gain of a time invariant linear system in the frequency domain which is utilised to analyse the stability of the system by considering the phase and gain margins [27]. Bode plots for each transfer function were determined analytically and numerically and their results were compared to assess the stability of the system.

3.1.1 Analytical solution

The following procedure required to determine the Bode plot analytically was implemented for each transfer function determined in Section 2.3. To demonstrate this procedure, the first transfer function is considered. Firstly, the transfer function was transformed to the frequency domain by replacing s with $i\omega$. The transfer function in the frequency domain is shown in Equation 3.1.

$$T(s) = -\frac{1}{m\omega^2} \tag{3.1}$$

The transfer function in Equation 3.1 was converted into decibel units whereby log laws were used to simplify the equation. This simplified equation is shown below.

$$m(\omega) = 20\log\left|\frac{1}{m}\right| - 40\log|\omega| \tag{3.2}$$

Based on the transfer function, it is evident that there are two common building blocks, these being constant gain, K, and multiple zeros at the origin. The magnitude of the constant gain in the current transfer function is $-\frac{1}{m}$. The phase angle for this constant gain is given in Equation 3.3 [28].

$$\emptyset(\omega) = -180° \text{ for } K < 0 \tag{3.3}$$

For multiple zeros at the origin, the phase angle is defined as [28]:

$$\emptyset(\omega) = n90° \text{ where } n = 2 \tag{3.4}$$

The phase portion of the bode plots remained the same for all four transfer functions for both common building blocks. This is because each transfer function had a common s^2 term in the denominator, and a negative constant gain term due to the presence of i^2. The magnitude of the gain of each transfer function differed based on the constant gain, however the shape of the bode plot did not change.

3.1.2 Numerical solution

The transfer functions obtained through the linearization process were passed through the MATLAB bode() function and the following plots were output:

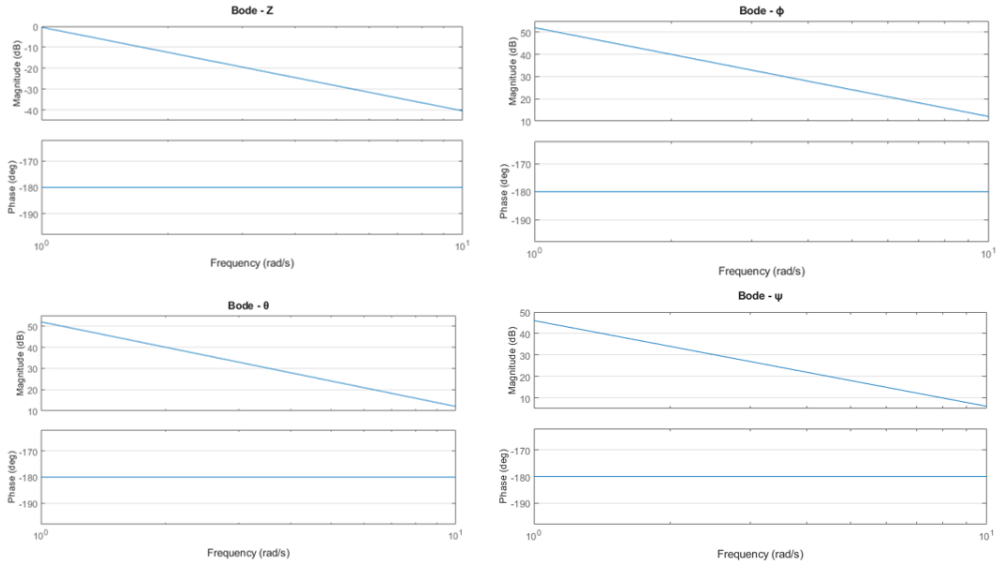


Figure 3: Bode Plots for each Transfer Function

3.1.3 Solution validation

The results obtained from the analytical and numerical bode plot creation show similarity. As discussed in Section 3.1.1, the phase graph will remain the same for all transfer functions. This is evident in Figure 3, whereby the phase graphs are all a horizontal line at $-180°$, which was the result obtained in the analytical analysis. Section 3.1.1 additionally states that the behaviour of the gain graphs would remain the same for all the transfer functions, however, the magnitude of the gain would change. Figure 3 shows that all four transfer functions have a gain graph which is a decreasing straight line. However, the maximum and minimum magnitudes differ for each transfer function, showing agreement with the analytical results.

The double integrator (s^2) nature of the system further explains this instability. In the frequency domain, the Bode plot's gain decreases as frequency increases, this leads to unbounded outputs for bounded inputs which violate the Bounded-Input, Bounded-Output (BIBO) stability criteria. The $-180°$ angle displays a complete phase lag that shows the output is out of phase with the input coupled with the gain not crossing 0 dB indicates that the system is on the verge of instability. There is no energy dissipation of the system due to the lack of damping, seen in the denominator having no additional s term, causing the results to diverge.

3.2 Nyquist Plot

3.2.1 Analytical solution

The following procedure required to determine the Nyquist plot analytically was implemented for each transfer function determined in Section 2.3. To demonstrate this procedure, the first transfer function is considered. The frequency transfer function was obtained by replacing s with $i\omega$, as shown in Equation 3.5.

$$T(s) = -\frac{1}{m\omega^2} \tag{3.5}$$

Based on Equation 3.5, it is evident that there is only a real portion of the equation. This indicates the Nyquist plot will exist only on the real axis. The value of the frequency transfer function when $\omega = 0$ and $\omega = \infty$ were determined to be $-\infty$ and 0 respectively. Therefore, the Nyquist plot of this transfer function approaches the origin from negative infinity and reaches the origin when $\omega = \infty$. When solving for the value of ω which causes the frequency transfer function to intersect with the real axis, a magnitude of ∞ is obtained, which further reinforces that the plot lies on the real axis. This result was obtained for all four transfer functions since the coefficient in front of the ω^2 term did not influence the final plot.

3.2.2 Numerical solution

The analytical analysis above is reinforced through the use of the Nyquist() function in MATLAB. The four transfer functions found through the linearized model were defined and the plots are seen below:

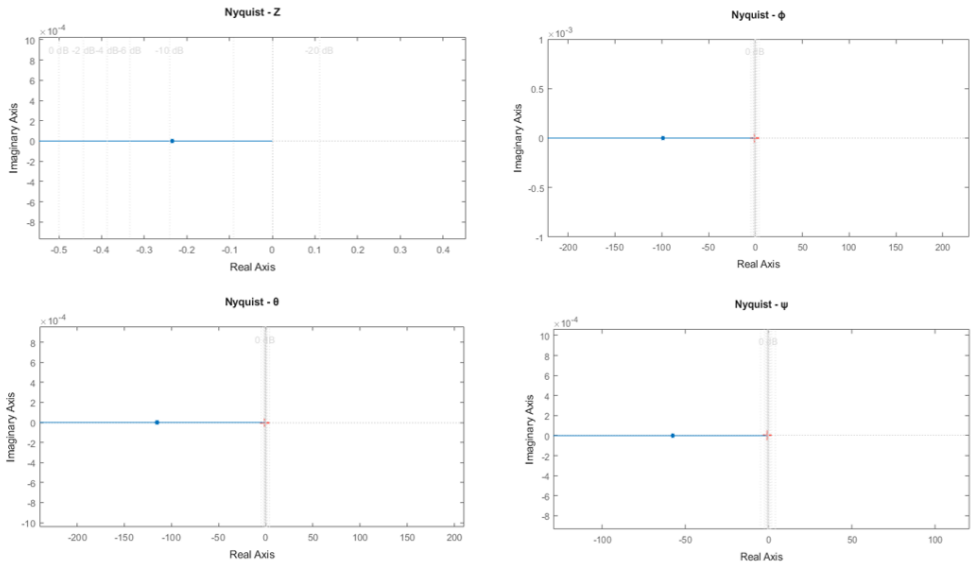


Figure 4: Nyquist Plots of the Four Transfer Functions

3.2.3 Solution validation

The Nyquist plots for the uncontrolled system's open-loop transfer functions exhibit the expected behaviour of systems with double integrators: the plots lie entirely along the negative real axis, running from $-\infty$ at low frequencies to 0 as frequency increases. This behaviour indicates a system with no damping and phase lag of 180

degrees (as the bode plot indicates), meaning the system is on the verge of instability in a feedback configuration. Although these plots do not technically encircle the critical point −1+0j, they pass directly over it, which would result in instability if closed-loop feedback were applied. This is further supported by the Routh-Hurwitz criterion, where the absence of some lower-order terms (due to integrators) confirms that the system is not internally stable.

3.3 Additional Numerical Stability Analysis

The stability of the drone is further examined through numerical simulations. The linear model was simulated using MATLAB linear tools and the stability was assessed for a step and impulse input of U_1, U_2, U_3 and U_4 as well as a selection of random U inputs. The results of these simulations are seen in Figure 5, 6 and 7 below:

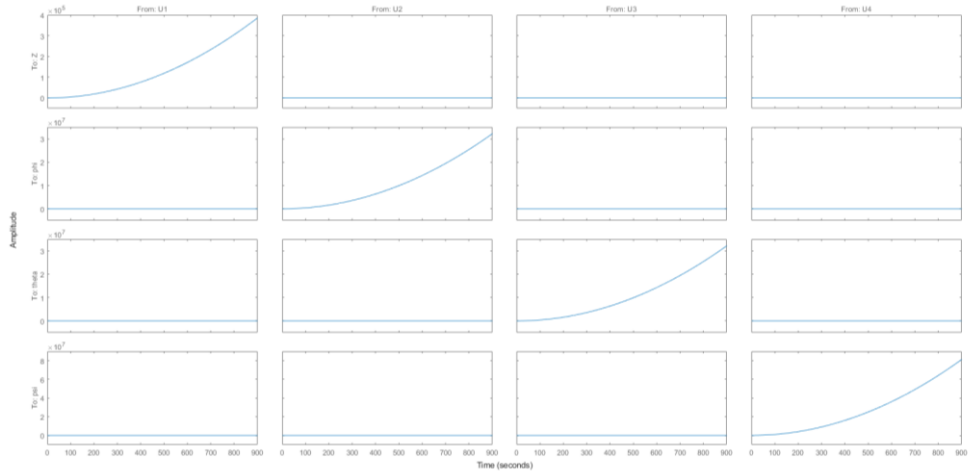


Figure 5: Uncontrolled Linear Step Response

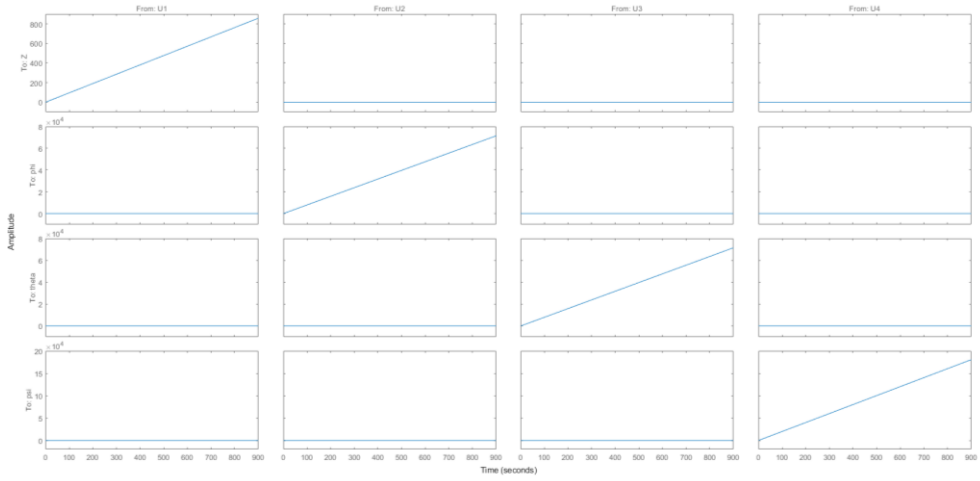


Figure 6: Uncontrolled Linear Impulse Response

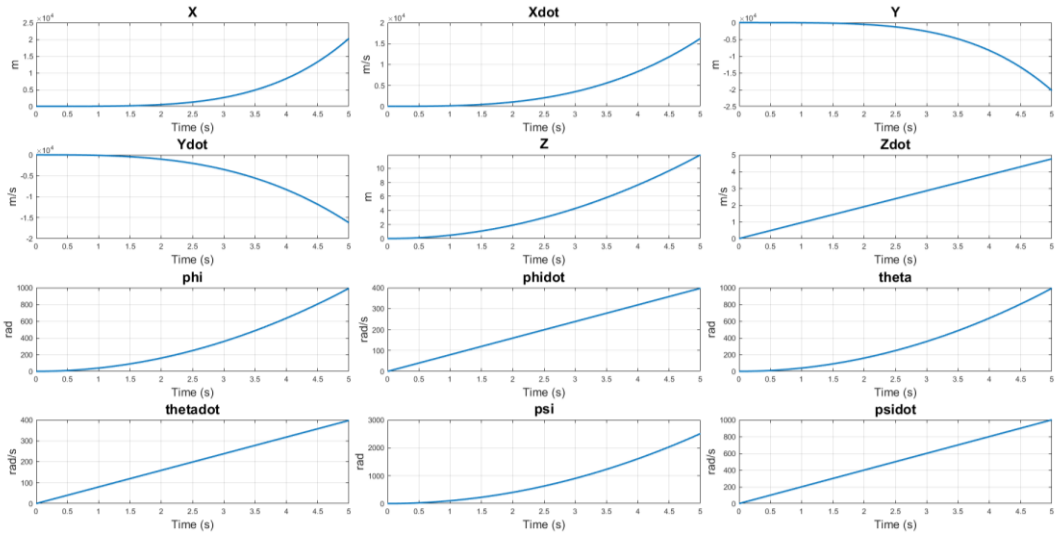


Figure 7: 12 State Response to Varied U Inputs

The above figures clearly show unstable behaviour for all cases. The drone remains stable for the singular case in which U_1 balances gravity but is highly unstable for any other input, illustrating the marginal stability of the double integrator systems and the unbounded outputs from bounded inputs that are expected for this system. These simulation results, as well as the in-depth root analyses in the previous section clearly indicated the need for a control system to allow the commanding of a hover and to meet the performance requirements a farming UAV needs.

4 PID DEVELOPMENT

4.1 Desired Roots

The optimal settling time and percentage overshoot were determined to be 1.2 seconds and 8% respectively. These values offer a good balance between responsiveness and stability. An 8% overshoot ensures that the system reacts quickly without excessive peaking. A 1.2 s settling time allows the system to stabilize within a short duration, making it suitable for real-time control while avoiding aggressive control effort. Given the farming drone scenario, precision higher than this is unnecessary.

Using the settling time and percentage overshoot, the desired damping ratio and natural frequency were determined, where:

$$\%\text{OS} = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \tag{4.1}$$

$$t_s = \frac{4}{\omega_n \zeta} \tag{4.2}$$

The transfer function is of second order and as such, the damping ratio, natural frequency and Equation 4.3 were used to determine the desired poles.

$$s_{1,2} = -\omega_n\zeta \pm i\omega_n\sqrt{1-\zeta^2} \tag{4.3}$$

The percentage overshoot and settling time remained constant across all four transfer functions. Due to this, only one set of desired poles was required for the entire system, which was of magnitude $s_{1,2\ desired} = -3.33 \pm 4.15i$.

The percentage overshoot and settling time remained constant across all four transfer functions. Due to this, only one set of desired poles was required for the entire system, which was of magnitude $s_{1,2\ desired} = -3.33 \pm 4.15i$.

4.2 Proportional Control

4.2.1 Analytical solution

The controller implementation was initially done using proportional control alone. The following analysis was performed on all three transfer functions; however, it is demonstrated using the altitude, Z, transfer function. The closed loop block diagram with unity feedback for altitude is seen below:

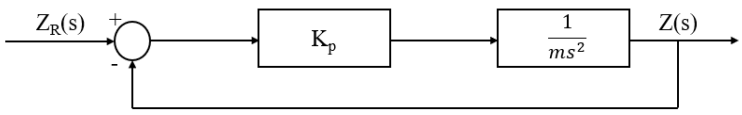


Figure 8: Block Diagram of the First Transfer Function with Proportional Control

The closed loop transfer function is thus,

$$G_p(s) = \frac{\frac{K_p}{m}}{s^2 + \frac{K_p}{m}} \tag{4.4}$$

Several conclusions about the stability of this system were drawn from assessing the characteristic polynomial:

$$D(s) = s^2 + \frac{K_p}{m} \tag{4.5}$$

Firstly, the Routh-Hurwitz criteria for stability were not met as the coefficient of the *s* term is 0. Moreover, the roots of this polynomial are:

$$s = \pm j\sqrt{\frac{K_p}{m}} \tag{4.6}$$

These are purely imaginary roots indicating marginal stability and BIBO instability. For the purposes of this time invariant linear system, this indicated instability of the control system. Beyond this, the 0 coefficient of the *s* term in the closed loop transfer function indicated no damping was present in this system and a sustained oscillation response to a disturbance was expected. A simple root locus plot of P-control confirms this, as evident in Figure 9.

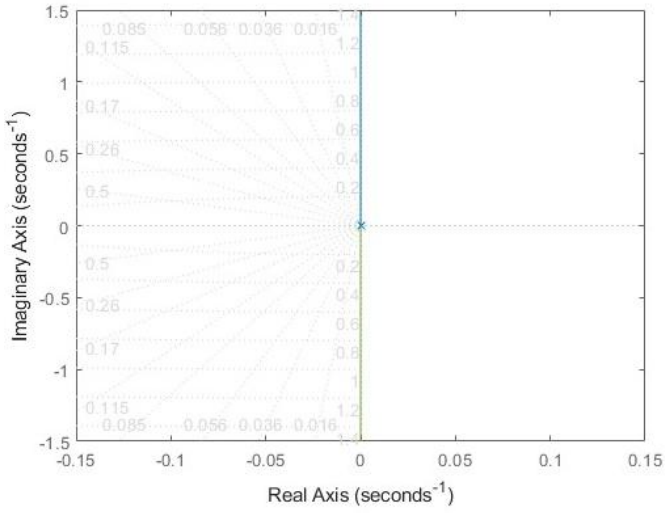


Figure 9: Root Locus Plot of the System with Proportional Control Only

As seen in Figure 9, the locus does not enter the LHP for any value of K_p meaning proportional gain cannot create stability for a double integrator system for any gain value.

An attempt to find the gain value for the design roots described in Section 4.1 was made using the root locus technique as follows:

$$1 + K_p P(s) = 0 \tag{4.7}$$

$$1 + \frac{K_p}{ms^2} = 0 \rightarrow K_p = -ms^2 \tag{4.8}$$

By substituting in the design point roots found in Section 4.1 for s above, an imaginary gain value that does not fall on a locus of the root locus plot is found, reaffirming that control is not possible for a P-control scheme. This is evident for all transfer functions since the variation in constant gain does not change the behaviour of the system with proportional control only.

4.2.2 Numerical solution

The analytical assessment of proportional control seen above is confirmed by the MATLAB numerical simulation. A number of K_p values were simulated with all values producing a similar, expected result. That result being undamped oscillation, as shown in Figure 10.

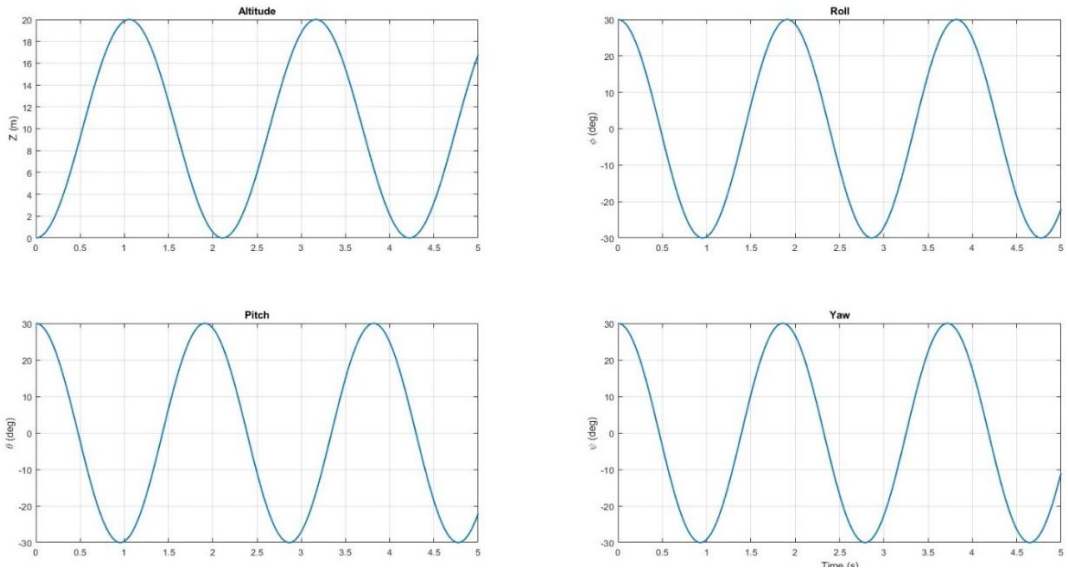


Figure 10: Numerical Results Obtained for Proportional Control Only

It is evident that proportional control alone is insufficient to stabilize the double integrator system, as it results in undamped, unstable behaviour. To improve stability, I-control was tested next.

4.3 Integral Control

4.3.1 Analytical solution

In a similar manner to the P-control, I-control was implemented by defining the controller transfer function as an Integral gain resulting in the following block diagram and closed loop transfer function:

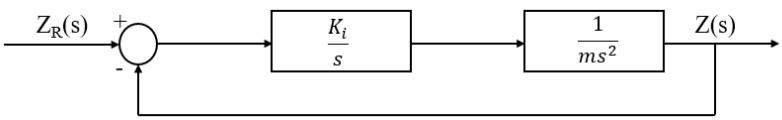


Figure 11: Block diagram of the First Transfer Function with Integral Control

$$G_i(s) = \frac{\frac{K_i}{m}}{s^3 + \frac{K_i}{m}} \tag{4.9}$$

Moreover, the open loop function for the system is a triple integrator:

$$G(s)C(s) = \frac{K_i}{ms^3} \tag{4.10}$$

The root locus plot for such a system is shown in Figure 12.

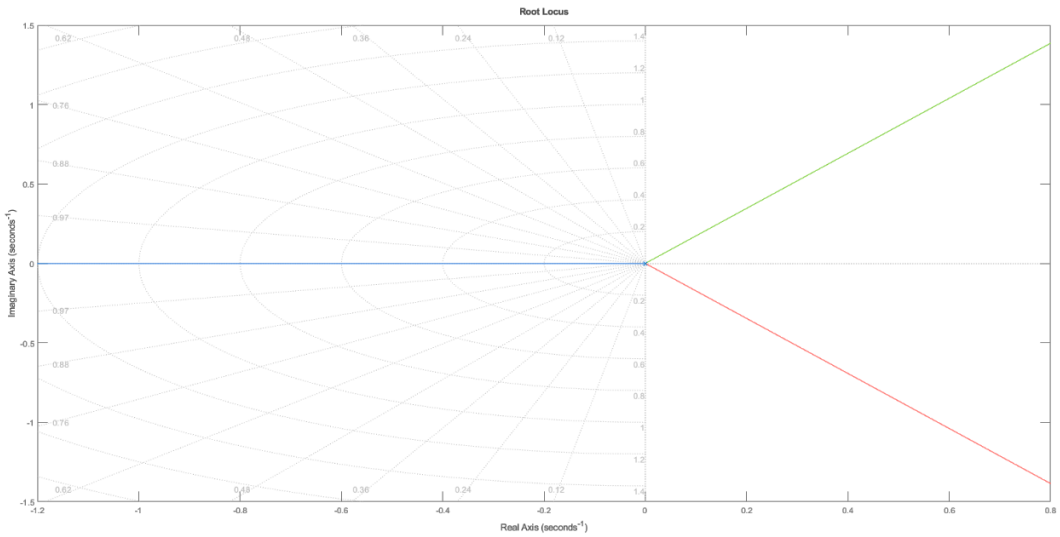


Figure 12: Root Locus Plot for I-Controlled Triple Integrator

From the above, it is clear that 3 poles exist at 0, with an increasing a K_i value causing a pole to move into the RHP. This indicates that, much like P-control, I-control alone cannot create system stability.

4.3.2 Numerical solution

The numerical simulation of an I-controlled system, shown in Figure 13, confirms the need for additional control terms.

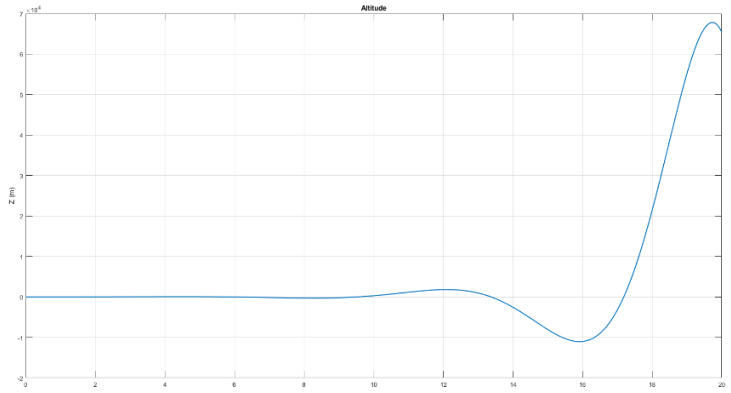


Figure 13: I-Controlled System

As evident in Figure 13, convergence is not achieved when utilising integral control alone. As such, PI control was considered, which is discussed in the following section.

4.4 Proportional – Integral Control

4.4.1 Analytical solution

Integral control was introduced to the control transfer function such that,

$$C(s) = K_p + \frac{K_i}{s} \tag{4.11}$$

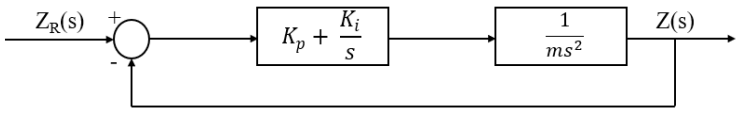


Figure 14: Block Diagram of the First Transfer Function with PI Control

This block diagram was reduced to the PI-control closed loop transfer function:

$$G_{PI}(s) = \frac{\frac{K_p}{m}s + K_i}{s^3 + \frac{K_p}{m}s + \frac{K_i}{m}} \tag{4.12}$$

For PI-control to result in stability, the real parts of the roots of the characteristic polynomial must be negative.

$$s^3 + \frac{K_p}{m}s + \frac{K_i}{m} \tag{4.13}$$

However, this expression lacks an s^2 term, violating the Routh–Hurwitz stability condition that all powers of s must have positive, non-zero coefficients. When constructing the Routh array, the presence of a zero in the s^2 row immediately implies instability. Therefore, the system was deemed unstable under PI control alone, regardless of the gain values. This implied a need for an additional damping term, such as derivative action, to achieve a stable response.

4.4.2 Numerical solution

The MATLAB implementation of the linear model was again used to confirm this instability. The behaviour of the first transfer function, Z, under PI control is shown in Figure 15.

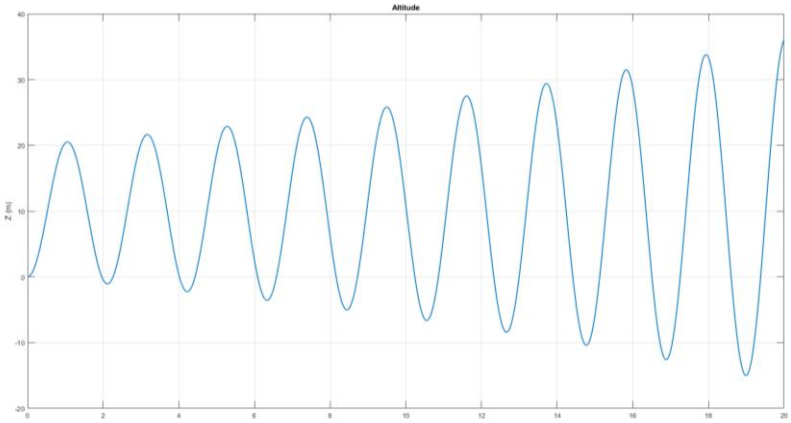


Figure 15: Numerical Solution of the First Transfer Function with Proportional Integral Control

Figure 15 shows continuous oscillatory behaviour whereby the amplitude is increasing with time. This proves that stability cannot be achieved with PI control alone as the behaviour of the function does not settle to an equilibrium point. Subsequently, PD control was considered, which is discussed below.

4.5 Proportional – Derivative Control

4.5.1 Analytical solution

The previous sections have clearly demonstrated the need for the implementation of damping into the closed loop transfer functions. By defining a derivate term in the control transfer function, damping was introduced to the close loop transfer function.

$$C(s) = K_p + K_d s \tag{4.14}$$

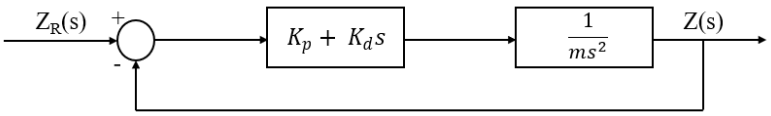


Figure 16: Block diagram of the First Transfer Function with PD Control

$$G_{PD}(s) = \frac{\frac{K_p}{m} + K_D s}{s^2 + \frac{K_D}{m} s + \frac{K_P}{m}} \tag{4.15}$$

The characteristic polynomial of the PD-controller is a second order polynomial that does not violate the Routh-Hurwitz stability criteria. To calculate the values of the gains that achieve the performance criteria specified in Section 4.1, the design point roots were used. Recall that $s_{1,2\ desired} = -3.33 \pm 4.15i$. The polynomial with these

roots is $s^2 + 6.66s + 28.3114$. By equating coefficients of these characteristic polynomials, the design point gains were found to be:

$$K_D = 6.993 \tag{4.16}$$

$$K_P = 29.7268 \tag{4.17}$$

A similar approach for the other 3 transfer functions was followed to determine all necessary PD-controller gains.

Table 2: K_p and K_d of Each Transfer Function

Transfer function	$\mathbf{K_p}$ and $\mathbf{K_d}$
$\frac{\mathbf{1}}{\boldsymbol{ms^2}}$	$K_D = 6.993$; $K_P = 29.7268$
$\frac{\mathbf{1}}{\boldsymbol{I_{xx}s^2}}$	$K_D = 0.0166$; $K_P = 0.0706$
$\frac{\mathbf{1}}{\boldsymbol{I_{yy}s^2}}$	$K_D = 0.0166$; $K_P = 0.0706$
$\frac{\mathbf{1}}{\boldsymbol{I_{zz}s^2}}$	$K_D = 0.0332$; $K_P = 0.1412$

4.5.2 Numerical solution

The results of the numerical simulation of the drone are shown below for a simple scenario of a commanded hover at 10 m from 0m and -30° attitudes. The response in all states shows classical second-order behaviour with mild overshoot and smooth settling within approximately 1.2 seconds as designed for. The %OS is slightly higher than the design point of 8% with a value of 10%.

Notably, the x and y positions display the expected coupling to the roll and pitch disturbances. The velocities in these directions converge accurately to 0 without prolonged oscillation indicating that a stable hover is achieved. However, the transient time during which these values are non-zero does mean that the x and y direction do drift, an issue that can only be controlled by an additional outer loop control scheme. Though, this controller does achieve a hover as required.

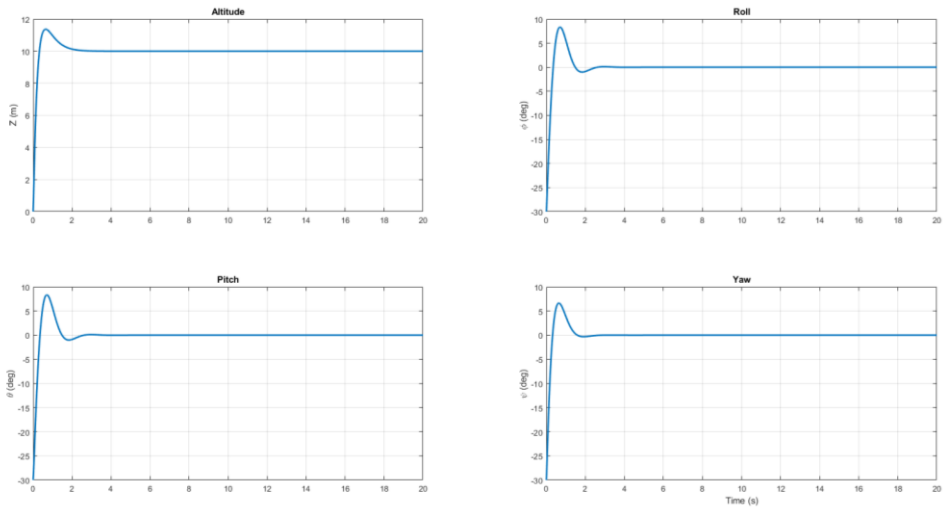


Figure 17: Simulation of Four Controlled States

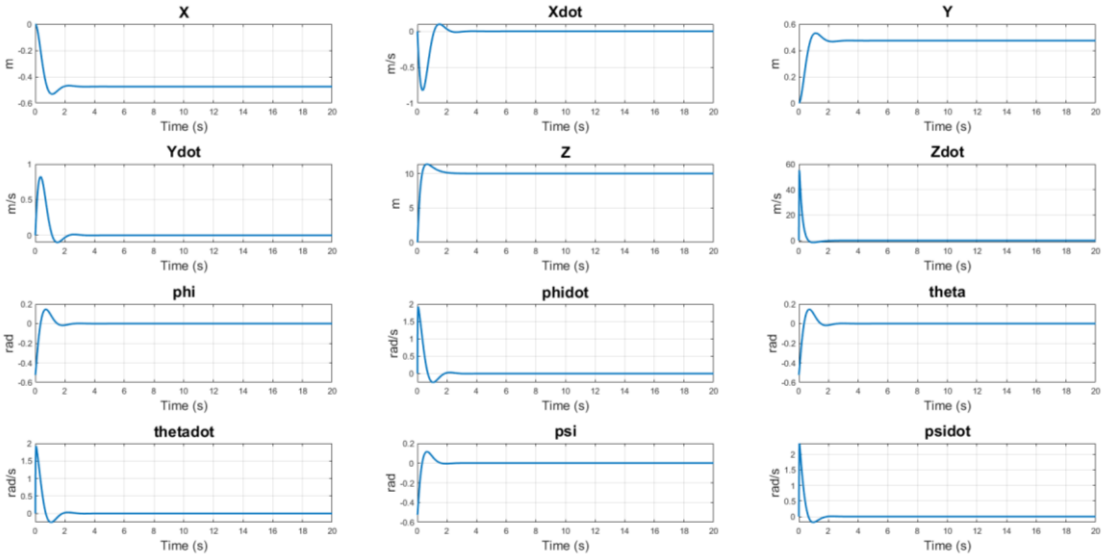


Figure 18: Simulation of All Twelve States with PD Control

4.6 PID Control

4.6.1 Analytical solution

Finally, PID control was implemented. A high amount of steady state error was not observed in the PD-control response, so integral control was not necessarily necessary. Moreover, the introduction of a new root through the s^3 term may reduce the efficacy of 2nd order solutions. Thus, a comparison between PID and PD is shown in Section 4.6.2 to decide on the best control scheme.

To determine the gains analytically for PID control, block diagrams for each transfer functions were first drawn. The block diagram of the first transfer function is shown in Figure 19.

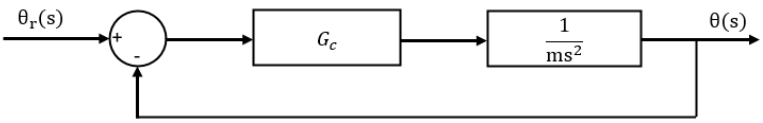


Figure 19: Block Diagram Including PID Control of the First Transfer Function

G_c in Figure 19 represents the PID controls to be implemented in the model and is defined as:

$$\mathrm{G_c} = \frac{K_{overall}(\mathrm{s}+\mathrm{a})(\mathrm{s}+\mathrm{b})}{\mathrm{s}} \tag{4.18}$$

Where a and b are constants to be determined and $\mathrm{K_{overall}}$ the overall constant gain. The difference between the sum of the zeroes' angles and the sum of the poles' angles must equal to 180°. This is shown in Equation 4.19 and was used to determine the value of a.

$$\sum \theta_Z - \sum \theta_P = -180° \tag{4.19}$$

A diagram of the angles to be considered is shown in Figure 20.

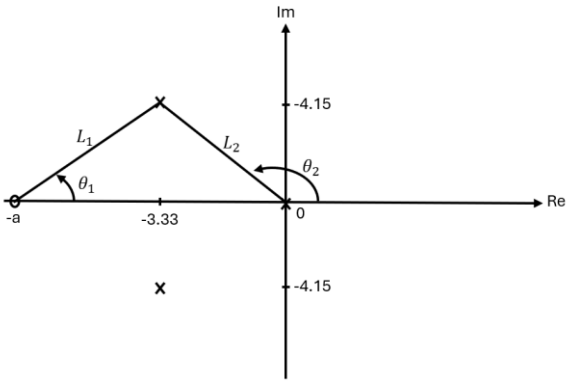


Figure 20: Plot Used to Calculate PID Controls

The zero angle, θ_1, is the angle between the line drawn from the undetermined point a to the desired root $-3.33 + 4.15i$ and the horizontal. The pole angle, θ_2, is the obtuse angle between the line drawn from the roots obtained from the characteristic equation, 0, to the desired root and the horizontal. Since the roots obtained from the characteristic equation are repeated, θ_2 must be accounted for twice in Equation 4.19. As such, the value of a was determined to be 4.35. Once the value of a was determined, it was necessary to determine the overall constant gain using Equation 4.20.

$$K = \frac{\prod L_P}{\prod L_Z} \tag{4.20}$$

Where the pole length was L_2 in Figure 20, which is the distance between the zeros of the characteristic equation and the desired root. Once again, this had to be accounted for twice since the roots obtained are repeated. The zero length was L_1 in Figure 20, which is the distance between the calculated point a and the desired root. Therefore, the value of $K_{overall}$ was determined to be 2.504. However, this value needed to be multiplied by the constant

gain for each transfer function to obtain the value of $K_{overall}$. Table 3 shows the final $K_{overall}$ values for each transfer function.

Table 3: $K_{overall}$ of each Transfer Function

Transfer function	$\mathbf{K_{overall}}$
$\frac{1}{ms^2}$	2,53
$\frac{1}{I_{xx}s^2}$	0,0063
$\frac{1}{I_{yy}s^2}$	0,0063
$\frac{1}{I_{zz}s^2}$	0,0125

The magnitude of b had to be assumed since its effect on the system could not be observed analytically. This value was assumed to be 0.1, which was deemed acceptable since it is close to zero, which ultimately results in a pole zero cancellation in the function G_c, which was required. However, this value was manually tuned in MATLAB when obtaining the numerical solution to determine the value which provided the most optimal results. Once all required inputs were determined for G_c, the different gains were extracted. The gains for each transfer function are shown in Table 4.

Table 4: Gain Values Obtained for each Transfer Function

Gain	$\frac{1}{ms^2}$	$\frac{1}{I_{xx}s^2}$	$\frac{1}{I_{yy}s^2}$	$\frac{1}{I_{zz}s^2}$
K_P	11.4	0.027	0.027	0.053
K_D	2.63	0,0063	0.0063	0.012
K_I	1.12	0.0027	0.0027	0.005

To ensure stability for each transfer function, the Routh-Hurwitz stability criterion was implemented, shown in Equation 4.21 [29]. The gains for each transfer function produced a value greater than zero and hence met the stability criterion.

$$K_D K_P - mK_I > 0 \tag{4.21}$$

4.6.2 Numerical solution

The tuned PID gains were applied to the same basic 10 m commanded hover simulation previously used for the PD-controlled system. While the overall response shape remained similar, several issues became apparent with the PID configuration. Notably, the introduction of integral control led to undershoot in the response and a longer settling time compared to the PD controller. These issues stem from two main factors: first, the addition of a third

pole means that second-order design assumptions no longer hold, requiring a degree of trial and error to find suitable, though imperfect, gains. Second, this additional pole influences system stability. Although it was placed far from the imaginary axis to minimize its impact, its presence still introduces third-order dynamics into an otherwise second-order system.

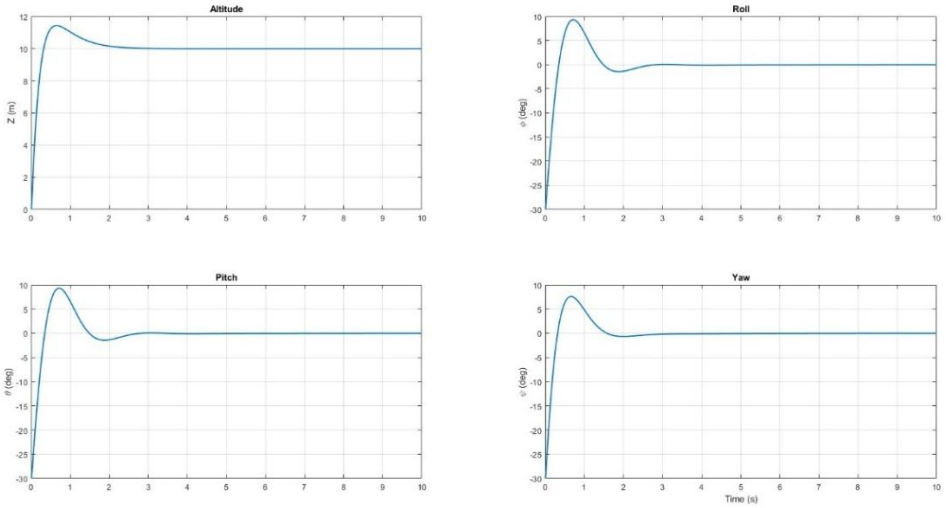


Figure 21: Simulation of 4 PID-Controlled States

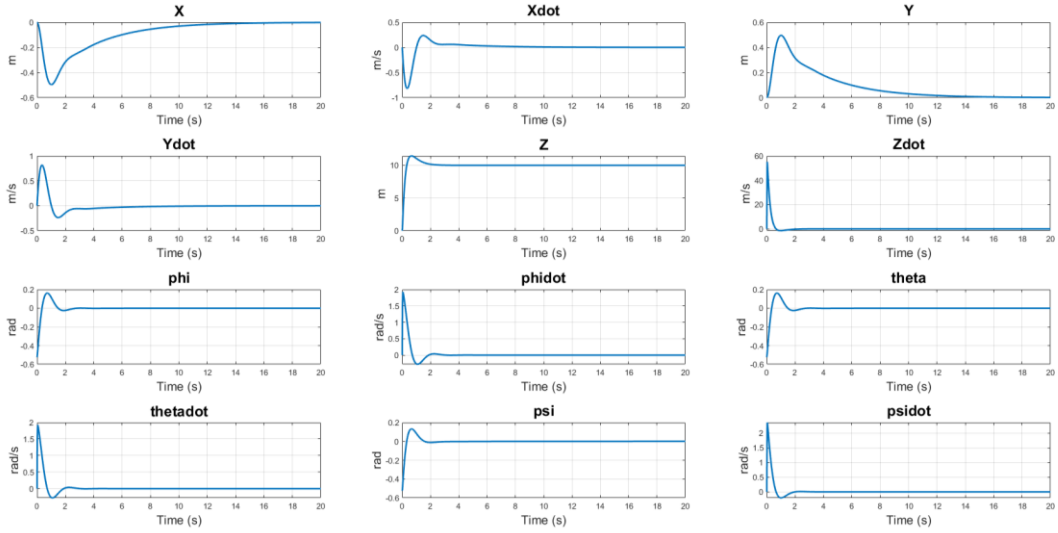


Figure 22: Full 12 State PID-Controlled Simulation

Clearly then, the introduction of an I-controller to the PD scheme is unnecessary at this stage of design as no steady state errors are present. Thus, the controller which will be used is the PD-control scheme.

5 SIMULINK ANALYSIS

5.1 Nonlinear Model

Nonlinear system analysis was initially performed in Simulink of the aforementioned equation of motion, discussed in Section 2.1.1. The nonlinear model utilised in this analysis is shown in Figure 23. Based on this model, the following plots, shown in Figure 24, were generated, each of which highlighting that the system is unstable under the influence of a disturbance.

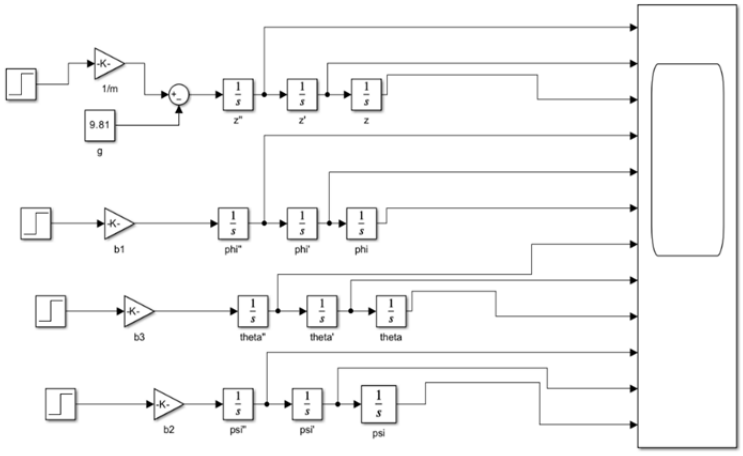


Figure 23: Non-Linear Simulink Model

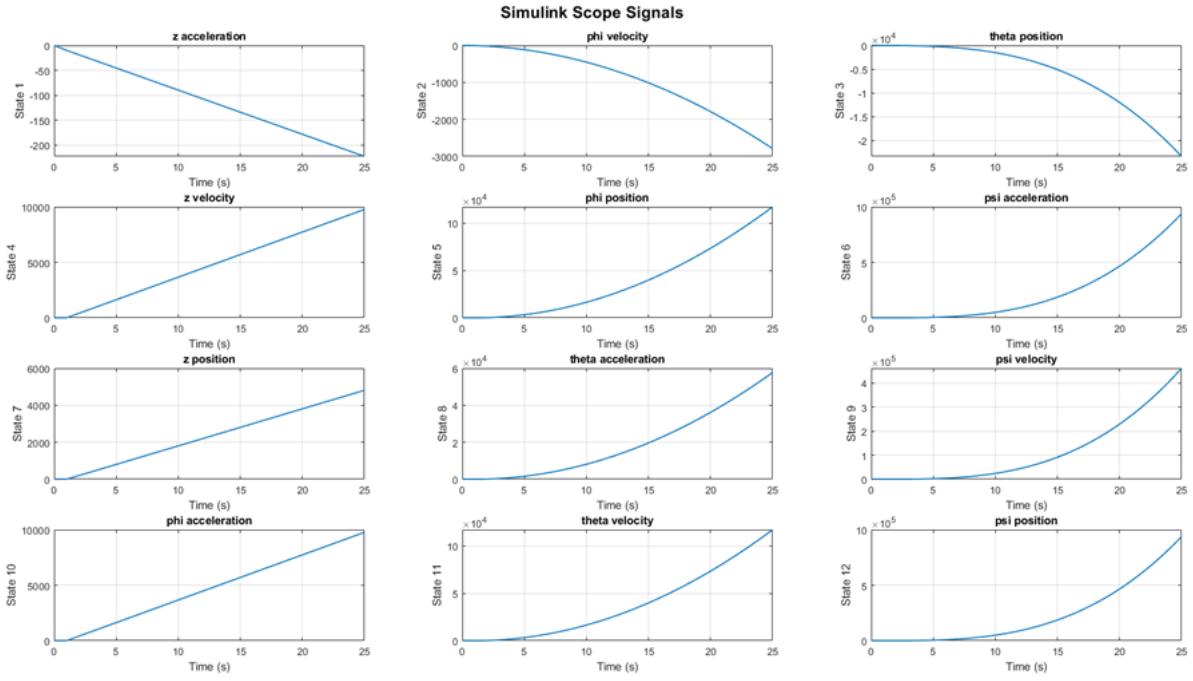


Figure 24: Non-Linear Response Behaviour of Drone under the Influence of a Step Input

The nonlinear system developed in Simulink was modified to include control, thereafter, being analysed under hover conditions. The control implemented was PID control whereby three gains were developed, namely an integrator, differentiator and saturation block. The architecture of the model with control is shown in Figure 25.

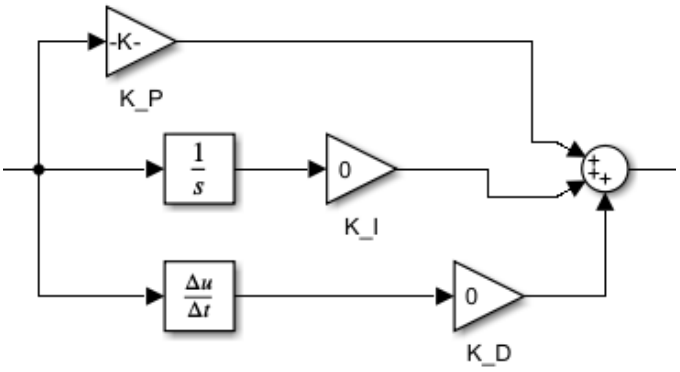


Figure 25: Classical PID Controller

The controller was created using the built-in Simulink PID block to avoid any potential errors which could arise through the filter coefficient that is auto implemented by the PID block. A controller was developed for the nonlinear system using the simplified governing equations of the quadrotor without small angle approximation being applied. This increased the analysis' complexity as it created coupling between the lateral term, y, and the bank angle. Thus, the full range of the non-linear motion of the drone was captured. To stabilize the drone, an inner and outer loop was developed in the controller model. The architecture of the non-linear model is shown in the figure below.

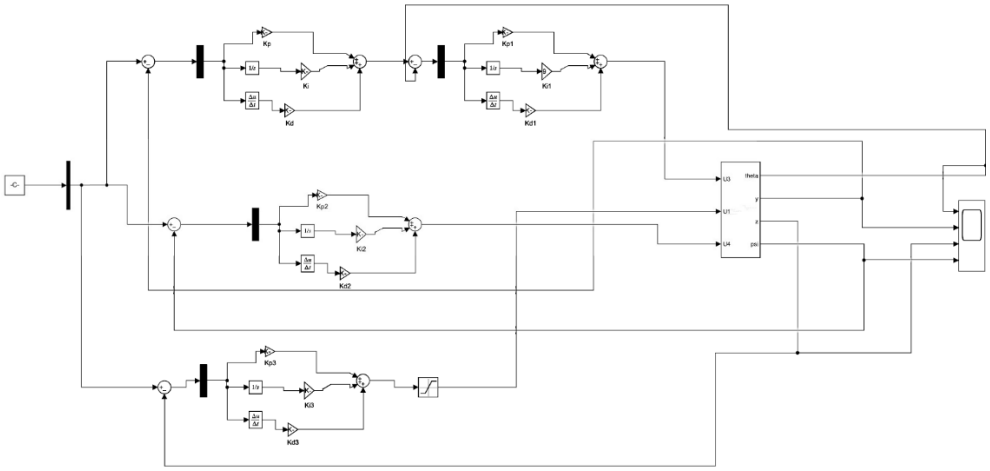


Figure 26: Non-linear controller architecture

The controller was found to be successful in stabilizing the drone in hover. The hover condition was simulated through the application of a constant block whose magnitude was equal to the weight of the drone. The controlled non-linear curves generated by the controller are shown below:

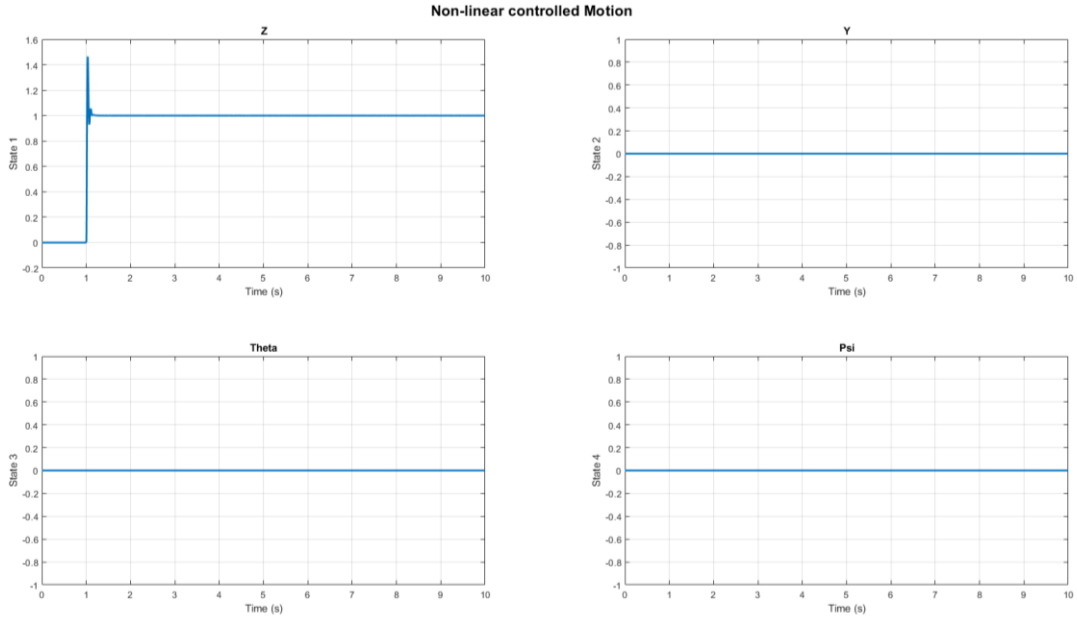


Figure 27: Non-linear hover graphs

The results are consistent with the behaviour of a typical hovering drone. There is no pitch, yaw or lateral motion from the time history curves. There is a slight fluctuation in the altitude time history curve whereby the sharp peaks and dips stem from the drone self-stabilizing after reaching its target altitude. However, this fluctuation dissipates within less than a second, which is ideal given the size of the drone. The fast response to the erratic motion in the vertical direction indicates that the controller is well designed for the nonlinear hover condition.

5.2 Comparison of Linear MATLAB model and Non-Linear Simulink Model

Both the linear and nonlinear models were subjected to a step input and the results were compared for all four states of interest under PID control. These plots are shown in Figure 28 and Figure 29 below.

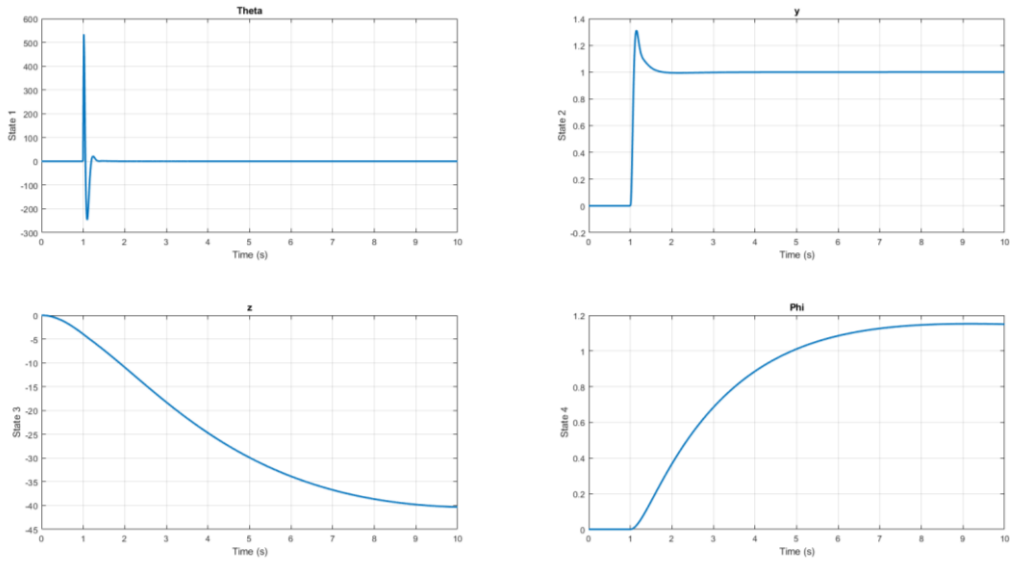


Figure 28: Response curves for non-linear PID control

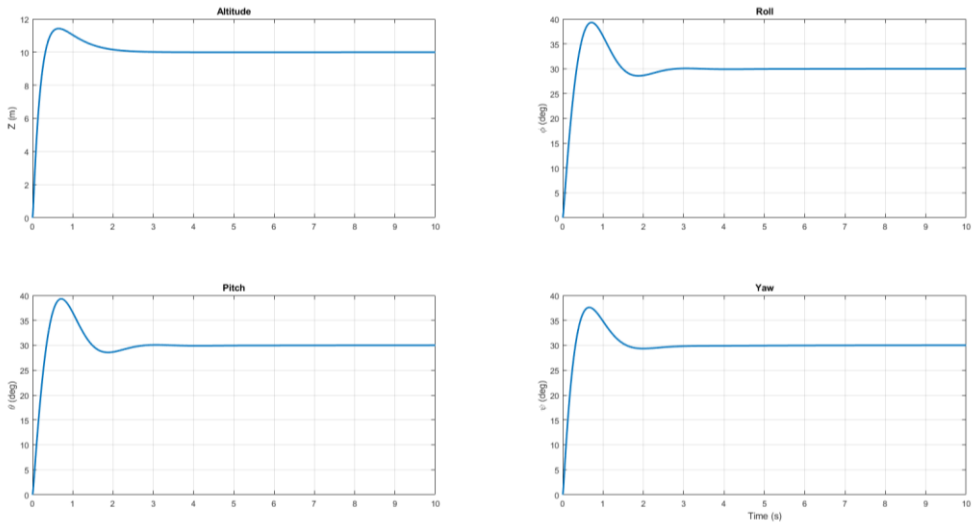


Figure 29: Response curves for linear PD control

Table 5: Comparison of linear MATLAB results with non-linear Simulink results

	Linear	**Non linear**
Altitude settle time	2 s	10 s
Altitude overshoot	15%	0%
Roll settle time	3 s	9 s
Roll overshoot	19%	0%
Pitch settle time	3 s	0.2 s
Pitch overshoot	19%	100 %
Yaw settle time	2.2 s	2 s
Yaw overshoot	16%	17 %

The comparison between the nonlinear and linear models reveals that the controlled states, altitude, roll, pitch, and yaw, settle significantly faster and with less overshoot in the linear case. This performance discrepancy arises because the controller was designed using linear control theory, assuming a linearized system model around an operating point. As a result, the linear PD-controller is optimally tuned for the simplified system, leading to ideal second-order responses with predictable dynamics.

In contrast, the nonlinear model captures additional complexities such as coupling effects and higher-order dynamics, which the linear controller is not explicitly designed to handle. Consequently, when applied to the nonlinear system, the controller exhibits reduced performance – illustrating the limitations of linear design when extended beyond its valid operating range. For this reason, the linear model is used to check robustness in the following section as it is a more accurate reflection of the controllers' capabilities in the region around the hover point.

6 MANOEUVRE IMPLEMENTATION

6.1 Freehand Throw

The first scenario in which the robustness of the designed controller that was tested was that of a freehand throw, such as in the case a farmer deploys the drone to begin the surveying process. The initial conditions of the drone were set in such a way to simulate this, with the goal of achieving hover at a 10 m altitude. The following initial conditions were used:

Table 6: Freehand Throw Initial Conditions

State	Initial Condition	Reason
$\boldsymbol{Z}$	1.2m	Arm height
$\dot{\boldsymbol{Z}}$	10m/s	Throw velocity
$\boldsymbol{\psi}$; $\boldsymbol{\theta}$; $\boldsymbol{\phi}$;	-π/6 to π/6 rad	Random initial angle
$\dot{\boldsymbol{\psi}}$; $\dot{\boldsymbol{\theta}}$; $\dot{\boldsymbol{\phi}}$	-π to π rad/s	Random initial rotational Velocity

The results for such a scenario are seen below, simulated over 20 seconds:

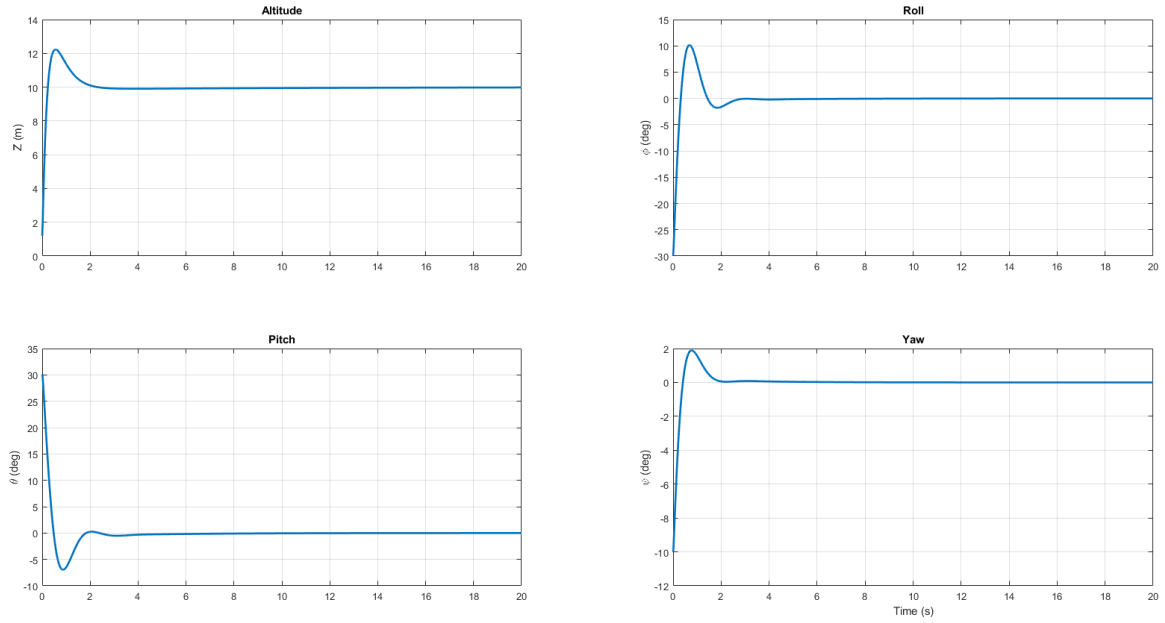


Figure 30: System Response to a Free Hand Throw

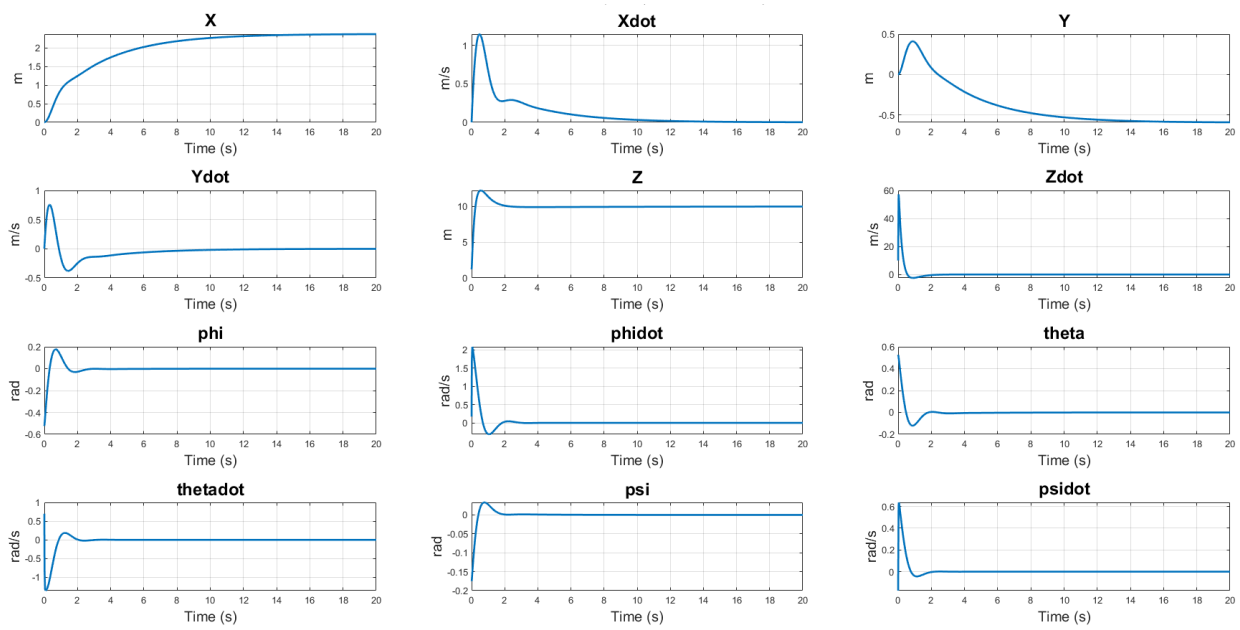


Figure 31: 12 State Responses to a Freehand Throw

The system is similar to a simple take-off scenario, though the initial rotational velocities to increase the overshoot for the attitude controllers. Again, the lack of outer loop control for the coupled x and y translations allows for these two states to drift away from 0. However, the other 10 states successfully reach a hover in a freehand throw scenario. In reality, this would simply mean the drone comes to a hover roughly 2 m away from the farmers initial x and y position at the required 10 m height. This is not detrimental to the performance of the drone and is to be expected without outer loop control. Thus, this scenario was deemed successfully controlled.

6.2 Free Fall

To simulate free fall, the drone's initial conditions were set as follows:

Table 7: Free fall Initial Conditions

State	Initial Condition	Reason
$\boldsymbol{Z}$	15m	Slightly higher than final 10m.
$\dot{\boldsymbol{Z}}$	-28m/s	Velocity after 40m fall.
$\boldsymbol{\psi}$; $\boldsymbol{\theta}$; $\boldsymbol{\phi}$;	-π to π rad	Random angles, incl. upside down
$\dot{\boldsymbol{\psi}}$; $\dot{\boldsymbol{\theta}}$; $\dot{\boldsymbol{\phi}}$	-2π to 2π rad/s	Spinning out of control

The simulation results are seen on the next page. The high initial velocities of both the Z and Roll states means that large overshoots are seen in these simulations. Critically however, the control was strong enough not only to recover from the spin-out but also settle at the required 10m before the drone impacted the ground at 0 m. Again, the couple coordinates exhibit some drift that is not detrimental to the hover performance. Thus, this control is deemed robust for this scenario.

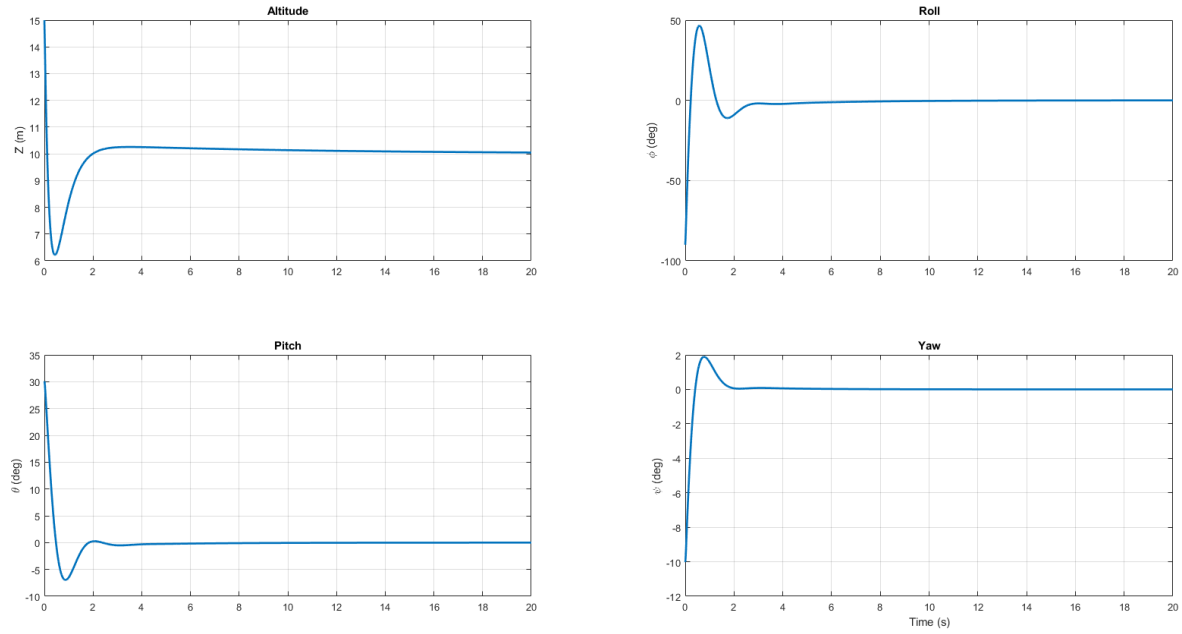


Figure 32: Free Fall Controlled States Simulation

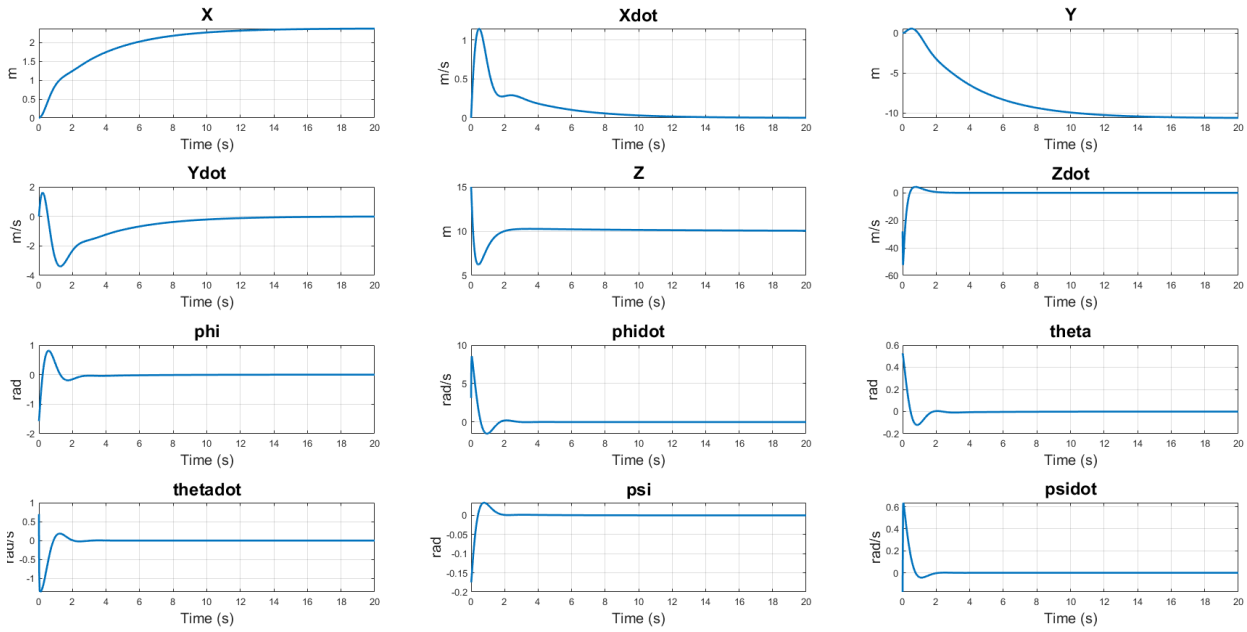


Figure 33: Free fall 12 State Simulation

6.3 Upside-Down Orientation

The upside-down orientation caused by the drone being caught on sugarcane or a similar deployment issue was simulated by simply setting the roll to -180°. The response is shown below:

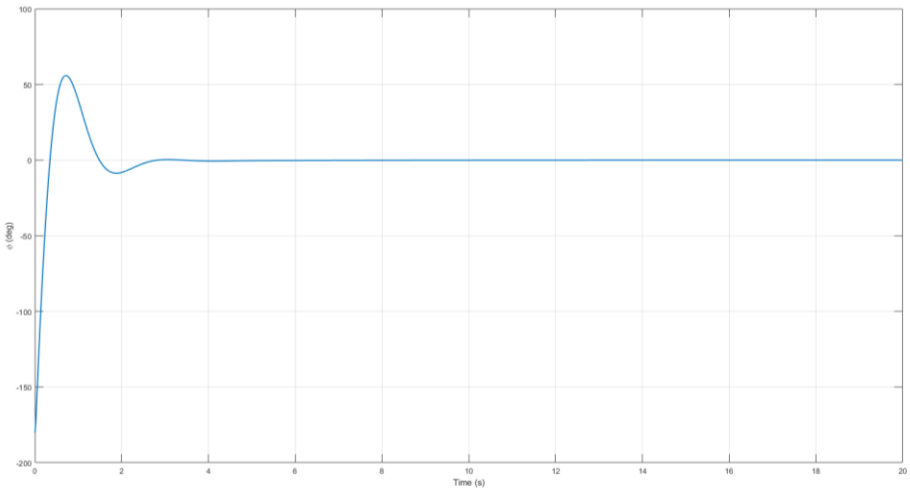


Figure 34: Upside Down Response

This scenario is effectively the same as commanding a hover and the PD-control scheme has no issue correcting an inverted position.

6.4 Aggressive Inputs

6.4.1 Wind Gusts

The first aggressive input unique to the farming UAV is that of a wind gust. The drone's initial conditions were set to represent a hover state that is suddenly exposed to strong lateral and rotational forces. This emulates an external disturbance while the drone attempts to hold position at 10 m with a horizontal attitude.

Table 8: Wind Gust Initial Conditions

State	Initial Condition	Reason
$\boldsymbol{Z}$	9.5m	Slightly blown down by gust.
$\boldsymbol{X}$	0 m	Initial Hover
$\dot{\boldsymbol{X}}$	5 m/s	Lateral Gust
$\boldsymbol{Y}$	0m	Initial Hover
$\dot{\boldsymbol{Y}}$	5 m/s	Lateral Gust
$\boldsymbol{\psi}$; $\boldsymbol{\theta}$; $\boldsymbol{\phi}$;	0 rad	Initial Hover
$\dot{\boldsymbol{\theta}}$; $\dot{\boldsymbol{\phi}}$	2π rad/s	Strong overturning moment.

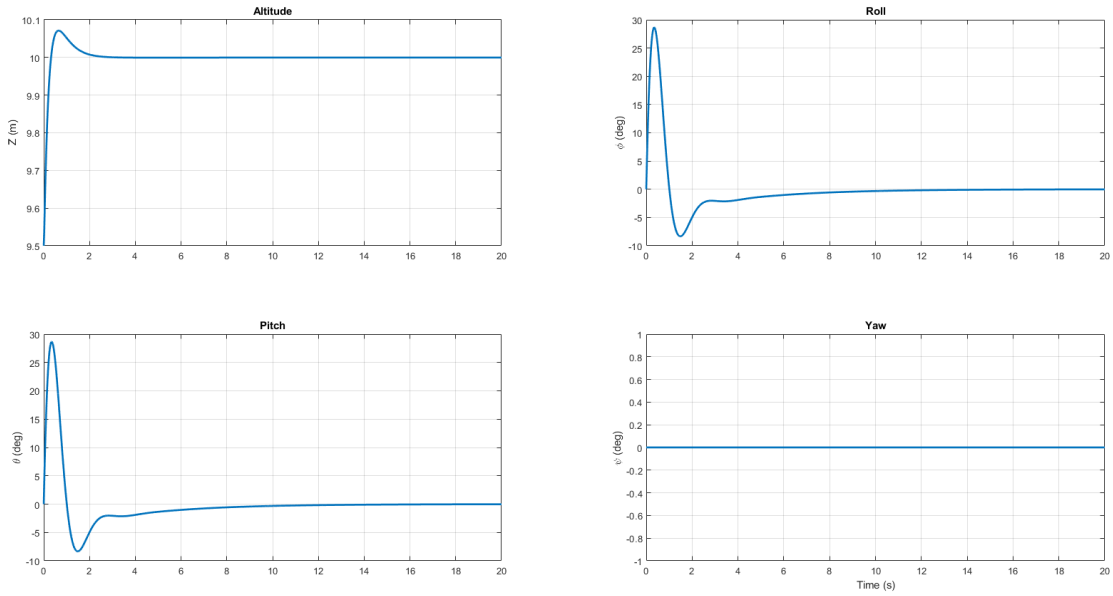


Figure 35: Wind Gust Controlled States for each Transfer Function

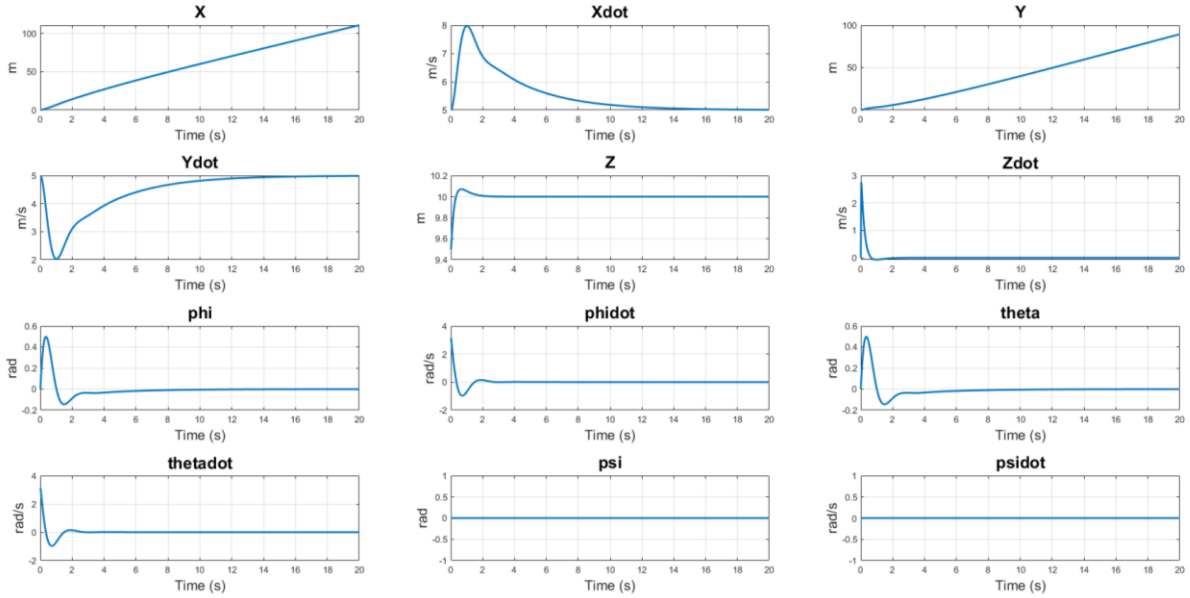


Figure 36: Wind Gust Controlled States

Whilst the controlled state responses are typical, a key issue with the control scheme and simulation emerged after a wind gust was applied. As already mentioned, the coupled states pertaining to lateral position and velocity are not directly associated with a controller. As such, the initial lateral velocities applied to the simulation have no feedback that reduces these inputs. This results in the initial conditions acting like a step response with an unbounded output as seen before in the uncontrolled scenario. In reality, the wind gust would die down, and the drone would ultimately return to a hover, albeit at a new lateral location, but the lack of an outer control loop represents a deficiency in the controller that aims to command a hover only.

6.4.2 Bird Strike

A bird striking the UAV is represented via a strong impulse to all 12 initial conditions. This differed from the step-like input of the wind gust as it was short-lived and of much higher magnitude.

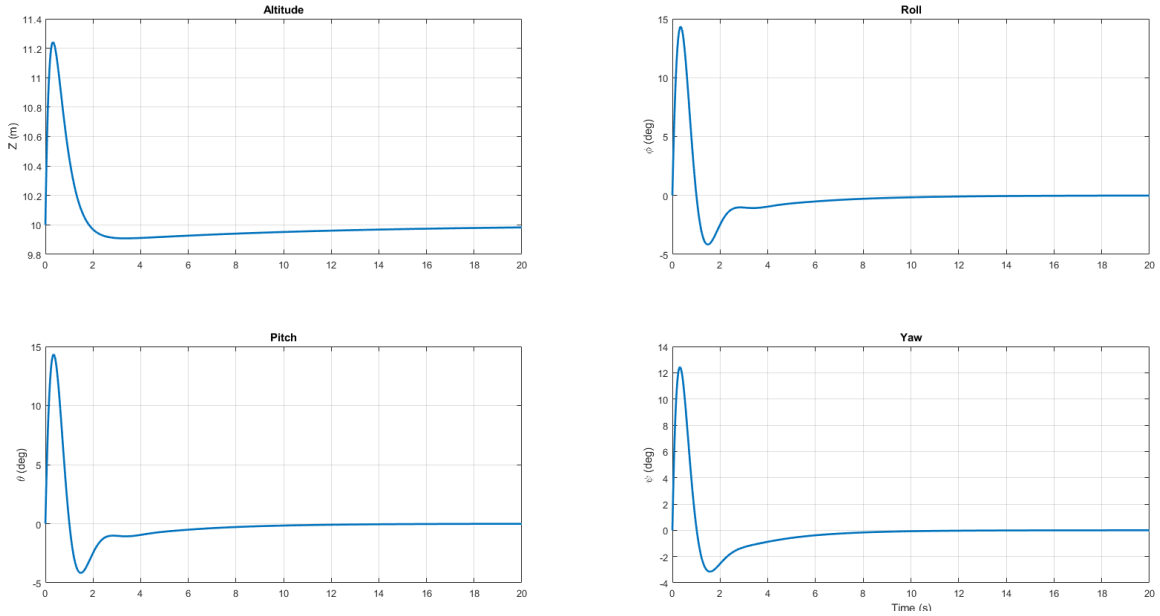


Figure 37: Bird Strike Impulse Controlled States

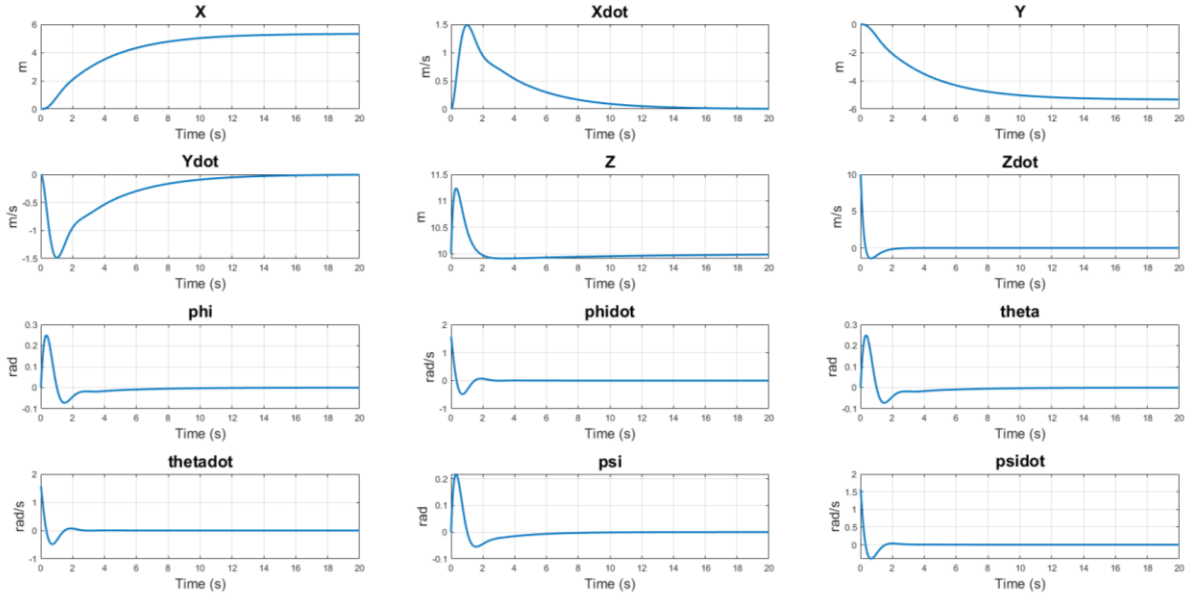


Figure 38: Bird Strike Impulse 12 State Simulation

The response to the bird strike shows the controller is highly robust in the face of strong impulses, as all states, which start at 0, are excited strongly and return to a steady hover in a precise and acceptable manner. Moreover, as expected, the uncontrolled states do return to a stable hover when the initial conditions are no longer applied, unlike in the gust state. This underlines that the limitation of a lack of outer control loop is only an issue when tracking of lateral positions is necessary.

7 PERFORMANCE ANALYSIS

7.1 Control Effort and Limitations

To evaluate the practicality and robustness of the controller, the control effort required to stabilize and maintain the commanded hover is examined. Specifically, the four control inputs U_1 to U_4, corresponding to thrust and rotational torques, are analysed over time. These signals represent the physical inputs applied to the quadrotor's motors and therefore determine the feasibility and efficiency of the control scheme.

The control signals generated by the controller are shown in Figure 39. These are computed using:

$$u(t) = C(s)e(t) \tag{6.1}$$

where *e(t)* is the tracking error between the reference signal and actual output. The outputs of the controller, the forces and moments U, are extracted and plotted below. This plot is the control effort for the highest demanding control scenario – free fall.

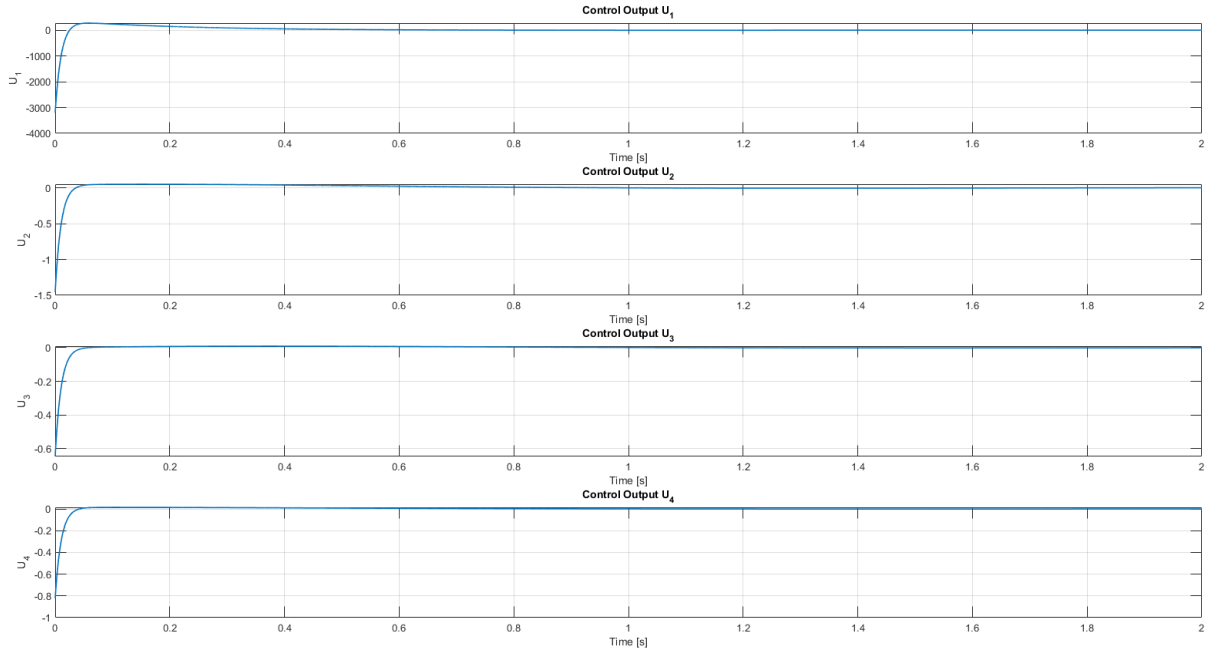


Figure 39: Control Effort

The control effort exhibits a strong initial correcting force/moment that quickly levels off at the 0 state. Importantly, the 0 state for U_1 is a thrust of 10.3 N as required to overcome gravity. Whilst the values of the control effort are not unreasonable, the rate at which the effort is made is slightly unrealistic. In reality, these forces would need to be applied over a slightly longer duration, and future implementations of the controller would be required to limit the maximum force output and rate of control effort.

That being said, the practicality of the controller can be further investigated by evaluating the total energy output of the controller using:

$$E = \int_0^T u(t)^T u(t) dt \tag{6.2}$$

This results in a total energy output of about 5500 J for the free fall scenario. Given the larger size of this drone and the intense free fall scenario, this value is seemingly reasonable and thus the control effort output is deemed practical, albeit in need of some limitations for real world implementation.

In addition to this, the aforementioned lack of an outer control loop for x and y control serve as another major controller limitation. Though, again, this limitation does not affect the required hover specifications and would only cause issues should lateral tracking be implemented.

Finally, the general limitations of a linearized model apply. Namely, the linearization about a hovering point assumes small deviations in position and orientation, which restricts the model's validity to conditions near equilibrium. As a result, the model may not accurately capture large-angle dynamics, nonlinear aerodynamic effects, actuator saturation, or coupling terms that may become significant during more aggressive manoeuvres or disturbances. This simplification is acceptable for control design and analysis around hover, but care must be taken when interpreting results beyond this region.

7.2 Control Instrumentation

The precision farming drone requires control instrumentation to achieve the manoeuvres and motion described in the preceding sections of this report. A table of the required instrumentation is shown in Table 9.

Table 9: Control Instrumentation Required [30]

Control Instrumentation	Purpose
Microcontroller	Functional control
Altimeter	Altitude tracking
Inertial Measurement Unit (IMU)	Orientation measurement
GPS module	Position measurement

The PID control system was developed on MATLAB and Simulink. The control model is applied to the drone by means of a microcontroller, such as a Raspberry Pi, whereby the code is exported to the microcontroller which then controls the various functions and manoeuvres of the drone.

One of the linearised equations of interest in the drone model is that for the altitude, Z, which provides the vertical height of the drone above the ground. This is required when performing manoeuvres such as hovering where this altitude is required to remain constant. To obtain the altitude measurement, an altimeter or barometer must be fitted to the drone seeing as their primary function is altitude tracking. The control is implemented on this data obtained to ensure altitude remains constant in the case of hovering.

The remaining linearised equations considered are those for the pitch, yaw and roll of the drone. When the drone experiences an aggressive manoeuvre, such as a bird strike, magnitudes of these states are required for the control to correct the drone's position back to its equilibrium point. These magnitudes can be obtained by fitting the drone with an IMU, which is responsible for the measurement of the drone's orientation.

Lastly, the drone is to be implemented in a precision farming context whereby it is to perform agricultural tasks on crop fields located outdoors. It is necessary for the farmer to be able to locate where the drone is at all times such that they can direct the drone controls. This is achievable with the implementation of a GPS model, which would provide the coordinate location of the drone, allowing the farmer to always access its position.

8 CONCLUSION

This report has demonstrated the substantial advancement in improving the reliability of agricultural drones within precision farming specifically the DJI Mavic 3M. It tackles the difficulties of stabilization when the drone experiences aggressive manoeuvres and disturbances – free throws, free falls, upside down orientations, bird strikes and wind gusts. The task of designing a control system to mitigate drone instability from these unpredictable conditions enable drones to be utilized by smallholder farmers in KwaZulu Natal, South Africa.

The methodology implemented captured the quadrotor's dynamics through mathematical modelling via nonlinear and linear representations. The nonlinear model is rooted in Newton-Euler principles and describe the drone's six degrees of freedom, while the linearized model simplifies the model for linear control techniques to be applied around the hover condition.

The models provided foundation for a detailed stability analysis, confirming the instability of the uncontrolled quadrotor system through Bode and Nyquist plots as the transfer functions of the models exhibited marginal stability due to double integrator behaviour. These results highlighted the need for active control to restore the drone to stable hovering. Thus, the exploration of various control strategies was implemented – proportional (P), integral (I), proportional-integral (PI), proportional-derivative (PD), and proportional-integral-derivative (PID) – to investigate which control method would provide optimal stability.

Performance criteria of a 1.2 second settling time with an overshoot of 8% were required, resulting in the PD controller showcasing favourable responses as stabilization with minimal oscillations was achieved across all aggressive manoeuvre scenarios. The controller successfully returned the commanded hover state of 10 metres. Furthermore, the drone's robustness displays that the drone will be capable of collecting high-resolution data in real-world environments where external disturbances like wind or collisions are common.

Despite obtaining a successful PD controller, a limitation that must be considered within the system is that the outer loop control for lateral tracking was not implemented in the controller design. While this did not compromise the hover state, it underscores a disconnect in the positioning of the drone in the scenario where a wind gust is experienced – as the drone will experience a positional drift in the x- and y-direction but keep the z- direction hover state. As a recommendation for future enhancements, precise navigation can be improved by administering outer loop controls to account for the lateral positioning. Additionally, validating the controller on a physical DJI Mavic 3M drone would aid in ensuring that the simulation and the hardware instrumentation can be calibrated to coincide.

In conclusion, this report has successfully developed and validated a PD control system that ensures stable hovering of an agricultural drone during aggressive manoeuvres. This robust control system enhances aerial surveillance and crop monitoring in precision agriculture. Moreover, directly supporting the indispensable use of drones within the Fourth Industrial Revolution to perform sustainable farming.

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